Model for "`Th1`"

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General Condition

- Basis type: `lgs`
- SAMB selection:
  - Type: `[Q, G]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_g, E_g, T_g, A_u, E_u, T_u]
  - Spin (s): `[0, 1]`
- Atomic selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_g, E_g, T_g, A_u, E_u, T_u]
  - Spin (s): `[0, 1]`
- Site-cluster selection:
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_g, E_g, T_g, A_u, E_u, T_u]
- Bond-cluster selection:
  - Type: `[Q, G, M, T]`
  - Rank: `[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]`
  - Irrep.: [A_g, E_g, T_g, A_u, E_u, T_u]
- Max. neighbor: `10`
- Search cell range: `(-2, 3), (-2, 3), (-2, 3)`
- Toroidal priority: `false`

Group and Unit Cell

- Group: SG No. 200 T_{h}^{1} $Pm\bar{3}$ [cubic]
- Associated point group: PG No. 200 T_{h} $m\bar{3}$ [cubic]
- Unit cell:
  $a =$ `1.00000`, $b =$ `1.00000`, $c =$ `1.00000`, $\alpha =$ `90.0`, $\beta =$ `90.0`, $\gamma =$ `90.0`
- Lattice vectors (conventional cell):
  $\boldsymbol{a}_1 =$ `[ 1.00000, 0.00000, 0.00000]`
  $\boldsymbol{a}_2 =$ `[ 0.00000, 1.00000, 0.00000]`
  $\boldsymbol{a}_3 =$ `[ 0.00000, 0.00000, 1.00000]`

Symmetry Operation

Table 1: Symmetry operation

#	SO	#	SO	#	SO	#	SO	#	SO
1	$\{1\|0\}$	2	$\{2_{001}\|0\}$	3	$\{2_{010}\|0\}$	4	$\{2_{100}\|0\}$	5	$\{3^{+}_{111}\|0\}$
6	$\{3^{+}_{-11-1}\|0\}$	7	$\{3^{+}_{1-1-1}\|0\}$	8	$\{3^{+}_{-1-11}\|0\}$	9	$\{3^{-}_{111}\|0\}$	10	$\{3^{-}_{1-1-1}\|0\}$
11	$\{3^{-}_{-1-11}\|0\}$	12	$\{3^{-}_{-11-1}\|0\}$	13	$\{-1\|0\}$	14	$\{\mathrm{m}_{001}\|0\}$	15	$\{\mathrm{m}_{010}\|0\}$
16	$\{\mathrm{m}_{100}\|0\}$	17	$\{-3^{+}_{111}\|0\}$	18	$\{-3^{+}_{-11-1}\|0\}$	19	$\{-3^{+}_{1-1-1}\|0\}$	20	$\{-3^{+}_{-1-11}\|0\}$
21	$\{-3^{-}_{111}\|0\}$	22	$\{-3^{-}_{1-1-1}\|0\}$	23	$\{-3^{-}_{-1-11}\|0\}$	24	$\{-3^{-}_{-11-1}\|0\}$		

Harmonics

Table 2: Harmonics

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
1	$\mathbb{Q}_0(A_g)$	A_g	0	Q, T	-	-	1
2	$\mathbb{G}_3(A_g)$	A_g	3	G, M	-	-	$\sqrt{15}xyz$
3	$\mathbb{Q}_4(A_g)$	A_g	4	Q, T	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$
4	$\mathbb{G}_0(A_u)$	A_u	0	G, M	-	-	1
5	$\mathbb{Q}_3(A_u)$	A_u	3	Q, T	-	-	$\sqrt{15}xyz$
6	$\mathbb{G}_4(A_u)$	A_u	4	G, M	-	-	$\frac{\sqrt{21}(x^4-3x^2y^2-3x^2z^2+y^4-3y^2z^2+z^4)}{6}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
7	$\mathbb{Q}_{2,1}(E_g)$	E_g	2	Q, T	-	1	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
8	$\mathbb{Q}_{2,2}(E_g)$					2	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
9	$\mathbb{Q}_{4,1}(E_g)$	E_g	4	Q, T	-	1	$-\frac{\sqrt{15}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{12}$
10	$\mathbb{Q}_{4,2}(E_g)$					2	$\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$
11	$\mathbb{G}_{2,1}(E_u)$	E_u	2	G, M	-	1	$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$
12	$\mathbb{G}_{2,2}(E_u)$					2	$\frac{\sqrt{3}(x-y)(x+y)}{2}$
13	$\mathbb{G}_{4,1}(E_u)$	E_u	4	G, M	-	1	$-\frac{\sqrt{15}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{12}$
14	$\mathbb{G}_{4,2}(E_u)$					2	$\frac{\sqrt{5}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{4}$
15	$\mathbb{Q}_{5,1}(E_u)$	E_u	5	Q, T	-	1	$\frac{3\sqrt{35}xyz(x-y)(x+y)}{2}$
16	$\mathbb{Q}_{5,2}(E_u)$					2	$\frac{\sqrt{105}xyz\left(x^2+y^2-2z^2\right)}{2}$
17	$\mathbb{G}_{1,1}(T_g)$	T_g	1	G, M	-	1	x
18	$\mathbb{G}_{1,2}(T_g)$					2	y
19	$\mathbb{G}_{1,3}(T_g)$					3	z
20	$\mathbb{Q}_{2,1}(T_g)$	T_g	2	Q, T	-	1	$\sqrt{3}yz$
21	$\mathbb{Q}_{2,2}(T_g)$					2	$\sqrt{3}xz$
22	$\mathbb{Q}_{2,3}(T_g)$					3	$\sqrt{3}xy$
23	$\mathbb{G}_{3,1}(T_g,1)$	T_g	3	G, M	1	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
24	$\mathbb{G}_{3,2}(T_g,1)$					2	$-\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$
25	$\mathbb{G}_{3,3}(T_g,1)$					3	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$
26	$\mathbb{G}_{3,1}(T_g,2)$	T_g	3	G, M	2	1	$\frac{\sqrt{15}x(y-z)(y+z)}{2}$
27	$\mathbb{G}_{3,2}(T_g,2)$					2	$-\frac{\sqrt{15}y(x-z)(x+z)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
28	$\mathbb{G}_{3,3}(T_g,2)$					3	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
29	$\mathbb{Q}_{4,1}(T_g,1)$	T_g	4	Q, T	1	1	$\frac{\sqrt{35}yz(y-z)(y+z)}{2}$
30	$\mathbb{Q}_{4,2}(T_g,1)$					2	$-\frac{\sqrt{35}xz(x-z)(x+z)}{2}$
31	$\mathbb{Q}_{4,3}(T_g,1)$					3	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
32	$\mathbb{Q}_{4,1}(T_g,2)$	T_g	4	Q, T	2	1	$\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$
33	$\mathbb{Q}_{4,2}(T_g,2)$					2	$-\frac{\sqrt{5}xz\left(x^2-6y^2+z^2\right)}{2}$
34	$\mathbb{Q}_{4,3}(T_g,2)$					3	$-\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$
35	$\mathbb{Q}_{1,1}(T_u)$	T_u	1	Q, T	-	1	x
36	$\mathbb{Q}_{1,2}(T_u)$					2	y
37	$\mathbb{Q}_{1,3}(T_u)$					3	z
38	$\mathbb{G}_{2,1}(T_u)$	T_u	2	G, M	-	1	$\sqrt{3}yz$
39	$\mathbb{G}_{2,2}(T_u)$					2	$\sqrt{3}xz$
40	$\mathbb{G}_{2,3}(T_u)$					3	$\sqrt{3}xy$
41	$\mathbb{Q}_{3,1}(T_u,1)$	T_u	3	Q, T	1	1	$\frac{x\left(2x^2-3y^2-3z^2\right)}{2}$
42	$\mathbb{Q}_{3,2}(T_u,1)$					2	$-\frac{y\left(3x^2-2y^2+3z^2\right)}{2}$
43	$\mathbb{Q}_{3,3}(T_u,1)$					3	$-\frac{z\left(3x^2+3y^2-2z^2\right)}{2}$
44	$\mathbb{Q}_{3,1}(T_u,2)$	T_u	3	Q, T	2	1	$\frac{\sqrt{15}x(y-z)(y+z)}{2}$
45	$\mathbb{Q}_{3,2}(T_u,2)$					2	$-\frac{\sqrt{15}y(x-z)(x+z)}{2}$
46	$\mathbb{Q}_{3,3}(T_u,2)$					3	$\frac{\sqrt{15}z(x-y)(x+y)}{2}$
47	$\mathbb{G}_{4,1}(T_u,1)$	T_u	4	G, M	1	1	$\frac{\sqrt{35}yz(y-z)(y+z)}{2}$
48	$\mathbb{G}_{4,2}(T_u,1)$					2	$-\frac{\sqrt{35}xz(x-z)(x+z)}{2}$

continued ...

Table 2

#	symbol	irrep.	rank	X	multiplicity	component	symmetry
49	$\mathbb{G}_{4,3}(T_u,1)$					3	$\frac{\sqrt{35}xy(x-y)(x+y)}{2}$
50	$\mathbb{G}_{4,1}(T_u,2)$	T_u	4	G, M	2	1	$\frac{\sqrt{5}yz\left(6x^2-y^2-z^2\right)}{2}$
51	$\mathbb{G}_{4,2}(T_u,2)$					2	$-\frac{\sqrt{5}xz\left(x^2-6y^2+z^2\right)}{2}$
52	$\mathbb{G}_{4,3}(T_u,2)$					3	$-\frac{\sqrt{5}xy\left(x^2+y^2-6z^2\right)}{2}$
53	$\mathbb{Q}_{5,1}(T_u,3)$	T_u	5	Q, T	3	1	$\frac{\sqrt{105}x(y-z)(y+z)\left(2x^2-y^2-z^2\right)}{4}$
54	$\mathbb{Q}_{5,2}(T_u,3)$					2	$\frac{\sqrt{105}y(x-z)(x+z)\left(x^2-2y^2+z^2\right)}{4}$
55	$\mathbb{Q}_{5,3}(T_u,3)$					3	$-\frac{\sqrt{105}z(x-y)(x+y)\left(x^2+y^2-2z^2\right)}{4}$

Basis in full matrix

Table 3: dimension = 24

#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)	#	orbital@atom(SL)
0	$\|s,\uparrow\rangle$ @A(1)	1	$\|s,\downarrow\rangle$ @A(1)	2	$\|p_x,\uparrow\rangle$ @A(1)	3	$\|p_x,\downarrow\rangle$ @A(1)	4	$\|p_y,\uparrow\rangle$ @A(1)
5	$\|p_y,\downarrow\rangle$ @A(1)	6	$\|p_z,\uparrow\rangle$ @A(1)	7	$\|p_z,\downarrow\rangle$ @A(1)	8	$\|s,\uparrow\rangle$ @A(2)	9	$\|s,\downarrow\rangle$ @A(2)
10	$\|p_x,\uparrow\rangle$ @A(2)	11	$\|p_x,\downarrow\rangle$ @A(2)	12	$\|p_y,\uparrow\rangle$ @A(2)	13	$\|p_y,\downarrow\rangle$ @A(2)	14	$\|p_z,\uparrow\rangle$ @A(2)
15	$\|p_z,\downarrow\rangle$ @A(2)	16	$\|s,\uparrow\rangle$ @A(3)	17	$\|s,\downarrow\rangle$ @A(3)	18	$\|p_x,\uparrow\rangle$ @A(3)	19	$\|p_x,\downarrow\rangle$ @A(3)
20	$\|p_y,\uparrow\rangle$ @A(3)	21	$\|p_y,\downarrow\rangle$ @A(3)	22	$\|p_z,\uparrow\rangle$ @A(3)	23	$\|p_z,\downarrow\rangle$ @A(3)		

Table 4: Atomic basis (orbital part only)

orbital	definition
$\lvert s\rangle$	1
$\lvert p_x\rangle$	x
$\lvert p_y\rangle$	y
$\lvert p_z\rangle$	z

SAMB: 1428 (all 1428)

- `A` : '`A`' site-cluster
  * bra: $\langle s,\uparrow\vert$, $\langle s,\downarrow\vert$
  * ket: $\vert s,\uparrow\rangle$, $\vert s,\downarrow\rangle$
  * wyckoff: `3d`

  $\boxed{\text{z1}}$ $\mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(s)}(A_g)$

  $\boxed{\text{z70}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g) = \dfrac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$

  $\boxed{\text{z71}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g) = \dfrac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

- `A` : '`A`' site-cluster
  * bra: $\langle s,\uparrow\vert$, $\langle s,\downarrow\vert$
  * ket: $\vert p_x,\uparrow\rangle$, $\vert p_x,\downarrow\rangle$, $\vert p_y,\uparrow\rangle$, $\vert p_y,\downarrow\rangle$, $\vert p_z,\uparrow\rangle$, $\vert p_z,\downarrow\rangle$
  * wyckoff: `3d`

  $\boxed{\text{z685}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_u) = \dfrac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2} - \dfrac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$

  $\boxed{\text{z686}}$ $\mathbb{G}_0^{(1,-1;c)}(A_u) = \dfrac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} + \dfrac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z687}}$ $\mathbb{G}_0^{(1,1;c)}(A_u) = \mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_0^{(s)}(A_g)$

$\boxed{\text{z750}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_g)}{2}$

$\boxed{\text{z751}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(s)}(A_g)}{2}$

$\boxed{\text{z752}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z753}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$

$\boxed{\text{z754}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$

$\boxed{\text{z755}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z880}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3}$

$\boxed{\text{z881}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3}$

$\boxed{\text{z882}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_0^{(s)}(A_g)}{3}$

$\boxed{\text{z883}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} + \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z884}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z885}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$

$\boxed{\text{z886}}$ $\mathbb{Q}_{3,1}^{(c)}(T_u, 2) = -\frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z887}}$ $$\mathbb{Q}_{3,2}^{(c)}(T_u,2) = \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$\boxed{\text{z888}}$ $$\mathbb{Q}_{3,3}^{(c)}(T_u,2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3}$$

$\boxed{\text{z889}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$\boxed{\text{z890}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$$

$\boxed{\text{z891}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3}$$

$\boxed{\text{z892}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} + \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$\boxed{\text{z893}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$\boxed{\text{z894}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$$

$\boxed{\text{z895}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$$

$\boxed{\text{z896}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$$

$\boxed{\text{z897}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$$

$\boxed{\text{z898}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} + \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$\boxed{\text{z899}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$\boxed{\text{z900}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$$

$\boxed{\text{z901}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z902}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2) = \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z903}}$ $\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3}$

$\boxed{\text{z904}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z905}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z906}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

- `A` : '`A`' site-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3d`

$\boxed{\text{z2}}$ $\mathbb{Q}_{0}^{(c)}(A_g,a) = \mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{0}^{(s)}(A_g)$

$\boxed{\text{z3}}$ $\mathbb{Q}_{0}^{(c)}(A_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z4}}$ $\mathbb{Q}_{0}^{(1,-1;c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z5}}$ $\mathbb{Q}_{0}^{(1,1;c)}(A_g) = \mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{0}^{(s)}(A_g)$

$\boxed{\text{z6}}$ $\mathbb{G}_{3}^{(c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$

$\boxed{\text{z7}}$ $\mathbb{G}_{3}^{(1,-1;c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$

$\boxed{\text{z72}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,a)=\frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$\boxed{\text{z73}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,a)=\frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$\boxed{\text{z74}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,b)=\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_g)}{2}$$

$\boxed{\text{z75}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,b)=\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_g)}{2}$$

$\boxed{\text{z76}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,c)=\frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}-\frac{\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$\boxed{\text{z77}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,c)=-\frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}-\frac{\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$\boxed{\text{z78}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,a)=\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_g)}{2}$$

$\boxed{\text{z79}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,a)=\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(s)}(A_g)}{2}$$

$\boxed{\text{z80}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,b)=\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}-\frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$\boxed{\text{z81}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,b)=-\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}-\frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$\boxed{\text{z82}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E_g)=\frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2}$$

$\boxed{\text{z83}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E_g)=\frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$$

$\boxed{\text{z208}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,a)=\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$$

$\boxed{\text{z209}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,a)=\frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$$

$\boxed{\text{z210}}$ $\mathbb{Q}_{2,3}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z211}}$ $\mathbb{Q}_{2,1}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} - \frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z212}}$ $\mathbb{Q}_{2,2}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} + \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z213}}$ $\mathbb{Q}_{2,3}^{(c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$

$\boxed{\text{z214}}$ $\mathbb{Q}_{4,1}^{(c)}(T_g,1) = -\frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z215}}$ $\mathbb{Q}_{4,2}^{(c)}(T_g,1) = \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z216}}$ $\mathbb{Q}_{4,3}^{(c)}(T_g,1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3}$

$\boxed{\text{z217}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z218}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z219}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z220}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} - \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z221}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z222}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$

$\boxed{\text{z223}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,1) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z224}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,1) = \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z225}}$ $\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3}$

$\boxed{\text{z226}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z227}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{6}$

$\boxed{\text{z228}}$ $\mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{3}$

$\boxed{\text{z229}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z230}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z231}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(s)}(A_g)}{3}$

$\boxed{\text{z232}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} + \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z233}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{6} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(s)}(E_g)}{2}$

$\boxed{\text{z234}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(s)}(E_g)}{3}$

- `A;A_001_1` : `'A'-'A'` bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `12a@12j`

  $\boxed{\text{z8}}$ $\mathbb{Q}_{0}^{(c)}(A_g) = \mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{0}^{(b)}(A_g)$

$\boxed{\text{z9}}$ $$\mathbb{Q}_{0}^{(1,-1;c)}(A_g) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z84}}$ $$\mathbb{G}_{0}^{(1,-1;c)}(A_u, a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a)}{3}$$

$\boxed{\text{z85}}$ $$\mathbb{G}_{0}^{(1,-1;c)}(A_u, b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b)}{3}$$

$\boxed{\text{z86}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z87}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z235}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z236}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$\boxed{\text{z237}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, a)}{3}$$

$\boxed{\text{z238}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$$

$\boxed{\text{z239}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u, b)}{3}$$

$\boxed{\text{z240}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$$

$\boxed{\text{z241}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z242}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z243}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z244}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,a) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z245}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,a) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z246}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z247}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z248}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z249}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z250}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z251}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z252}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z688}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z689}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z756}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$\boxed{\text{z757}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z758}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,c) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z759}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z907}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z908}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z909}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z910}}$ $$\mathbb{Q}_{3,1}^{(c)}(T_u,1) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z911}}$ $$\mathbb{Q}_{3,2}^{(c)}(T_u,1) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z912}}$ $$\mathbb{Q}_{3,3}^{(c)}(T_u,1) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z913}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z914}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z915}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z916}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z917}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u,b) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z918}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z919}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z920}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z921}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z922}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z923}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z924}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

- `A;A_001_1` : 'A'-'A' bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `12a@12j`

$\boxed{\text{z10}}$ $\mathbb{Q}_0^{(c)}(A_g,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z11}}$ $\mathbb{Q}_0^{(c)}(A_g,b) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$

$\boxed{\text{z12}}$ $\mathbb{Q}_0^{(c)}(A_g,c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$

$\boxed{\text{z13}}$ $\mathbb{Q}_4^{(c)}(A_g) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$

$\boxed{\text{z14}}$ $\mathbb{Q}_0^{(1,0;c)}(A_g,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z15}}$ $\mathbb{Q}_0^{(1,0;c)}(A_g,b) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$

$\boxed{\text{z16}}$ $\mathbb{Q}_0^{(1,0;c)}(A_g,c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$

$\boxed{\text{z17}}$ $$\mathbb{Q}_{4}^{(1,0;c)}(A_g) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z18}}$ $$\mathbb{G}_{3}^{(1,-1;c)}(A_g,a) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z19}}$ $$\mathbb{G}_{3}^{(1,-1;c)}(A_g,b) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3} - \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z20}}$ $$\mathbb{G}_{3}^{(1,-1;c)}(A_g,c) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z21}}$ $$\mathbb{G}_{3}^{(1,-1;c)}(A_g,d) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z88}}$ $$\mathbb{Q}_{3}^{(c)}(A_u) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z89}}$ $$\mathbb{Q}_{3}^{(1,-1;c)}(A_u,a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z90}}$ $$\mathbb{Q}_{3}^{(1,-1;c)}(A_u,b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z91}}$ $$\mathbb{Q}_{3}^{(1,-1;c)}(A_u,c) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z92}}$ $$\mathbb{Q}_{3}^{(1,0;c)}(A_u) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z93}}$ $$\mathbb{G}_{0}^{(c)}(A_u) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z94}}$ $$\mathbb{G}_{0}^{(1,-1;c)}(A_u,a) = \frac{\sqrt{5}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{5} + \frac{\sqrt{5}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{5}$$

$\boxed{\text{z95}}$ $$\mathbb{G}_{0}^{(1,-1;c)}(A_u,b) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z96}}$ $$\mathbb{G}_{4}^{(1,-1;c)}(A_u) = \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} - \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{30}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15}$$

$\boxed{\text{z97}}$ $\mathbb{G}_0^{(1,0;c)}(A_u) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$

$\boxed{\text{z98}}$ $\mathbb{G}_0^{(1,1;c)}(A_u, a) = \mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z99}}$ $\mathbb{G}_0^{(1,1;c)}(A_u, b) = \mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_0^{(b)}(A_g)$

$\boxed{\text{z100}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g, a) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z101}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g, a) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z102}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{6} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{6} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{3}$

$\boxed{\text{z103}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g, b) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$

$\boxed{\text{z104}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g, c) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, a)}{3}$

$\boxed{\text{z105}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g, c) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, a)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, a)}{2}$

$\boxed{\text{z106}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g, d) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u, b)}{3}$

$\boxed{\text{z107}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g, d) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u, b)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u, b)}{2}$

$\boxed{\text{z108}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z109}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z110}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{2}$

$\boxed{\text{z111}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,b) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$

$\boxed{\text{z253}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,c) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{2} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{2}$

$\boxed{\text{z254}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,c) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$

$\boxed{\text{z255}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,d) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{2} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{2}$

$\boxed{\text{z256}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,d) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$

$\boxed{\text{z257}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,a) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z258}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,a) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z259}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} + \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$

$\boxed{\text{z260}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,b) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{2}$

$\boxed{\text{z261}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,c) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$

$\boxed{\text{z262}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,c) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{2}$

$\boxed{\text{z263}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,d) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$

$\boxed{\text{z264}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,d) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{2}$

$\boxed{\text{z265}}$ $\mathbb{G}_{2,1}^{(c)}(E_u,a) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{2} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{2}$

$\boxed{\text{z266}}$ $\mathbb{G}_{2,2}^{(c)}(E_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$

$\boxed{\text{z267}}$ $\mathbb{G}_{2,1}^{(c)}(E_u,b) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$

$\boxed{\text{z268}}$ $\mathbb{G}_{2,2}^{(c)}(E_u,b) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$

$\boxed{\text{z269}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$

$\boxed{\text{z270}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$

$\boxed{\text{z271}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,b) = \frac{\sqrt{7}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{7} + \frac{\sqrt{7}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{7} + \frac{\sqrt{7}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{7}$

$\boxed{\text{z272}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,b) = -\frac{\sqrt{7}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{7} - \frac{\sqrt{21}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{7} + \frac{\sqrt{21}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14}$

$\boxed{\text{z273}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,c) = \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{0}^{(b)}(A_g)}{2}$

$\boxed{\text{z274}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,c) = \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{0}^{(b)}(A_g)}{2}$

$\boxed{\text{z275}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,d) = \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z276}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,d) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z277}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,e) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{2} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$

$\boxed{\text{z278}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,e) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$

$\boxed{\text{z279}}$ $\mathbb{G}_{4,1}^{(1,-1;c)}(E_u) = \frac{\sqrt{21}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14} - \frac{\sqrt{21}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21} - \frac{\sqrt{21}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} - \frac{\sqrt{21}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21} + \frac{2\sqrt{21}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21}$

$\boxed{\text{z280}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(E_u) = -\frac{\sqrt{21}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} + \frac{\sqrt{7}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{7} - \frac{\sqrt{21}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14} - \frac{\sqrt{7}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{7}$$

$\boxed{\text{z281}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E_u, a) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{2} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{2}$$

$\boxed{\text{z282}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z283}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E_u, b) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z284}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E_u, b) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$\boxed{\text{z285}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z286}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z287}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u, b) = \frac{\sqrt{2}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z288}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u, b) = \frac{\sqrt{2}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z289}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z290}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z291}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z292}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g, b) = -\frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u, 1)}{6}$$

$\boxed{\text{z293}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g, b) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u, 1)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u, 1)}{6}$$

$\boxed{\text{z294}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,b) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z295}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z296}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z297}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z298}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,d) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z299}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,d) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z300}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,d) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z301}}$ $$\mathbb{Q}_{4,1}^{(c)}(T_g,1) = -\frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z302}}$ $$\mathbb{Q}_{4,2}^{(c)}(T_g,1) = \frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z303}}$ $$\mathbb{Q}_{4,3}^{(c)}(T_g,1) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z304}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z305}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,a) = -\frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z306}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,a) = -\frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z307}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{18} - \frac{2\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{9} + \frac{2\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{9}$$

$\boxed{\text{z308}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} + \frac{2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{9} - \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{18} - \frac{2\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{9}$$

$\boxed{\text{z309}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,b) = -\frac{2\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{9} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{9} + \frac{2\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{9}$$

$\boxed{\text{z310}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z311}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,c) = -\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z312}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,c) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z313}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,d) = \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z314}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z315}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z316}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z317}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z318}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z319}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,2) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3} + \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{9} - \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{18} + \frac{\sqrt{6}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{18}$$

$\boxed{\text{z320}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,2) = -\frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3} + \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{18} + \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{9} - \frac{\sqrt{6}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{18}$$

$\boxed{\text{z321}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,2) = -\frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{18} - \frac{2\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{9} + \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{18}$$

$\boxed{\text{z322}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g,a) = \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z323}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g,a) = \frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z324}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_g,a) = \frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z325}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z326}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z327}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z328}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z329}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z330}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z331}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g,d) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z332}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g,d) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z333}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_g,d) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z334}}$ $$\mathbb{Q}_{4,1}^{(1,0;c)}(T_g,1) = -\frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z335}}$ $$\mathbb{Q}_{4,2}^{(1,0;c)}(T_g,1) = \frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z336}}$ $\mathbb{Q}_{4,3}^{(1,0;c)}(T_g,1) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$

$\boxed{\text{z337}}$ $\mathbb{G}_{1,1}^{(c)}(T_g,a) = \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z338}}$ $\mathbb{G}_{1,2}^{(c)}(T_g,a) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z339}}$ $\mathbb{G}_{1,3}^{(c)}(T_g,a) = \frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z340}}$ $\mathbb{G}_{1,1}^{(c)}(T_g,b) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z341}}$ $\mathbb{G}_{1,2}^{(c)}(T_g,b) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z342}}$ $\mathbb{G}_{1,3}^{(c)}(T_g,b) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z343}}$ $\mathbb{G}_{1,1}^{(c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z344}}$ $\mathbb{G}_{1,2}^{(c)}(T_g,c) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z345}}$ $\mathbb{G}_{1,3}^{(c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z346}}$ $\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,a) = -\frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10}$

$\boxed{\text{z347}}$ $\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,a) = -\frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{10} - \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10}$

$\boxed{\text{z348}}$ $\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{15} + \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10}$

$\boxed{\text{z349}}$ $\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{2}$

$\boxed{\text{z350}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{2}$$

$\boxed{\text{z351}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z352}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,c) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10}$$

$\boxed{\text{z353}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,c) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{10} - \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10}$$

$\boxed{\text{z354}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,c) = \frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{15} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10}$$

$\boxed{\text{z355}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10}$$

$\boxed{\text{z356}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{10} - \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10}$$

$\boxed{\text{z357}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,d) = \frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{15} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10}$$

$\boxed{\text{z358}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1a) = -\frac{\sqrt{5}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{15}$$

$\boxed{\text{z359}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1a) = -\frac{\sqrt{5}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z360}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1a) = \frac{\sqrt{5}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z690}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1b) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{15}$$

$\boxed{\text{z691}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1b) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{15}$$

$\boxed{\text{z692}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1b) = \frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{5} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{15}$$

$\boxed{\text{z693}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1c) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{15}$$

$\boxed{\text{z694}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1c) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{15}$$

$\boxed{\text{z695}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1c) = \frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{5} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{15}$$

$\boxed{\text{z696}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2a) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z697}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2a) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z698}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2a) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z699}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2b) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3} + \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z700}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2b) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z701}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2b) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z760}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2c) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3} + \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z761}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2c) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z762}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2c) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z763}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_g,a) = \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z764}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_g,a) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z765}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,a) = \frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z766}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,b) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z767}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z768}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,b) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z769}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z770}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,c) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z771}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z772}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$

$\boxed{\text{z773}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$

$\boxed{\text{z774}}$ $\mathbb{G}_{1,3}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z775}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3}$

$\boxed{\text{z776}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3}$

$\boxed{\text{z777}}$ $\mathbb{G}_{1,3}^{(1,1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$

$\boxed{\text{z778}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(T_g,c) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3}$

$\boxed{\text{z779}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(T_g,c) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3}$

$\boxed{\text{z780}}$ $\mathbb{G}_{1,3}^{(1,1;c)}(T_g,c) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$

$\boxed{\text{z781}}$ $\mathbb{G}_{3,1}^{(1,1;c)}(T_g,1) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3}$

$\boxed{\text{z782}}$ $\mathbb{G}_{3,2}^{(1,1;c)}(T_g,1) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3}$

$\boxed{\text{z783}}$ $\mathbb{G}_{3,3}^{(1,1;c)}(T_g,1) = \frac{\sqrt{3}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$

$\boxed{\text{z925}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z926}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z927}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z928}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u,b) = -\frac{\sqrt{30}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{30} + \frac{\sqrt{10}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10}$

$\boxed{\text{z929}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u,b) = \frac{\sqrt{10}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{10} - \frac{\sqrt{30}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{30} - \frac{\sqrt{10}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10}$

$\boxed{\text{z930}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u,b) = \frac{\sqrt{10}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10} + \frac{\sqrt{30}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{15}$

$\boxed{\text{z931}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u,c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z932}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u,c) = \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z933}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u,c) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z934}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u,d) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z935}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u,d) = -\frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z936}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u,d) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$\boxed{\text{z937}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u,e) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z938}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u,e) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z939}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u,e) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z940}}$ $$\mathbb{Q}_{3,1}^{(c)}(T_u,1) = -\frac{\sqrt{5}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} + \frac{\sqrt{15}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} - \frac{\sqrt{15}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15}$$

$\boxed{\text{z941}}$ $$\mathbb{Q}_{3,2}^{(c)}(T_u,1) = -\frac{\sqrt{15}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} - \frac{\sqrt{5}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} - \frac{\sqrt{15}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} - \frac{\sqrt{15}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z942}}$ $$\mathbb{Q}_{3,3}^{(c)}(T_u,1) = -\frac{\sqrt{15}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{5}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{5}$$

$\boxed{\text{z943}}$ $$\mathbb{Q}_{3,1}^{(c)}(T_u,2a) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3} - \frac{\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z944}}$ $$\mathbb{Q}_{3,2}^{(c)}(T_u,2a) = -\frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z945}}$ $$\mathbb{Q}_{3,3}^{(c)}(T_u,2a) = \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3} + \frac{\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z946}}$ $$\mathbb{Q}_{3,1}^{(c)}(T_u,2b) = -\frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z947}}$ $$\mathbb{Q}_{3,2}^{(c)}(T_u,2b) = \frac{\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z948}}$ $$\mathbb{Q}_{3,3}^{(c)}(T_u,2b) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z949}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} + \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30}$$

$\boxed{\text{z950}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10} + \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30} - \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30} - \frac{\sqrt{30}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30}$$

$\boxed{\text{z951}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{15}$$

$\boxed{\text{z952}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u,b) = \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z953}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u,b) = -\frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z954}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z955}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u,c) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{10}$$

$\boxed{\text{z956}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u,c) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{10} - \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{10}$$

$\boxed{\text{z957}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u,c) = \frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{15} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{10}$$

$\boxed{\text{z958}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1a) = \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{20} - \frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{20} - \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{60} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{60} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15}$$

$\boxed{\text{z959}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1a) = -\frac{\sqrt{10}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{20} + \frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{60} + \frac{\sqrt{10}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{20} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{60} - \frac{\sqrt{30}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z960}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1a) = -\frac{\sqrt{30}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30}$$

$\boxed{\text{z961}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1b) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{10} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{15}$$

$\boxed{\text{z962}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1b) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z963}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1b) = \frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{5} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z964}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2a) = \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{12} - \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{12} + \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{4} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{4}$$

$\boxed{\text{z965}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2a) = \frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{12} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{4} - \frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{12} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{4}$$

$\boxed{\text{z966}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2a) = -\frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6}$$

$\boxed{\text{z967}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2b) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z968}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2b) = -\frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z969}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2b) = \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$\boxed{\text{z970}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2c) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3} + \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z971}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2c) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z972}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2c) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{3}$

$\boxed{\text{z973}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z974}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z975}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z976}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{30}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{30} + \frac{\sqrt{10}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10}$

$\boxed{\text{z977}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,b) = \frac{\sqrt{10}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{10} - \frac{\sqrt{30}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{30} - \frac{\sqrt{10}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10}$

$\boxed{\text{z978}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,b) = \frac{\sqrt{10}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10} + \frac{\sqrt{30}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{15}$

$\boxed{\text{z979}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,c) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z980}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,c) = \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z981}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,c) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z982}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,d) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z983}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,d) = -\frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z984}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,d) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z985}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,e) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$

$\boxed{\text{z986}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,e) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z987}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,e) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z988}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{5}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} + \frac{\sqrt{15}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} - \frac{\sqrt{15}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15}$$

$\boxed{\text{z989}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{15}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} - \frac{\sqrt{5}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} - \frac{\sqrt{15}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} - \frac{\sqrt{15}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z990}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{15}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{5}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{5}$$

$\boxed{\text{z991}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2a) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3} - \frac{\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z992}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2a) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z993}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2a) = \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3} + \frac{\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z994}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2b) = -\frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z995}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2b) = \frac{\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z996}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2b) = \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z997}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z998}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z999}}$ $$\mathbb{Q}_{1,3}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z1000}}$ $$\mathbb{G}_{2,1}^{(c)}(T_u,a) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1001}}$ $$\mathbb{G}_{2,2}^{(c)}(T_u,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1002}}$ $$\mathbb{G}_{2,3}^{(c)}(T_u,a) = -\frac{\sqrt{2}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z1003}}$ $$\mathbb{G}_{2,1}^{(c)}(T_u,b) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1004}}$ $$\mathbb{G}_{2,2}^{(c)}(T_u,b) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1005}}$ $$\mathbb{G}_{2,3}^{(c)}(T_u,b) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1006}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z1007}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z1008}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z1009}}$ $$\begin{aligned}\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,b) = {} & \frac{\sqrt{42}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{42} + \frac{\sqrt{42}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{42} - \frac{\sqrt{14}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} \\ & - \frac{\sqrt{14}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14}\end{aligned}$$

$\boxed{\text{z1010}}$ $$\begin{aligned}\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,b) = {} & \frac{\sqrt{42}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{42} + \frac{\sqrt{14}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} \\ & + \frac{\sqrt{42}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{42} + \frac{\sqrt{14}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} + \frac{\sqrt{14}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14}\end{aligned}$$

$\boxed{\text{z1011}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,b) = -\frac{\sqrt{42}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21} + \frac{\sqrt{14}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{42}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{21}$$

$\boxed{\text{z1012}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,c)=\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z1013}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,c)=\frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z1014}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,c)=\frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z1015}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,d)=\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}+\frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}-\frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6}+\frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1016}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,d)=-\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}+\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6}+\frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}-\frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1017}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,d)=-\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}-\frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}+\frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1018}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,1)=-\frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{4}-\frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{4}-\frac{\sqrt{6}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{12}-\frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{12}$$

$\boxed{\text{z1019}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,1)=\frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{4}-\frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{12}+\frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{4}-\frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{12}$$

$\boxed{\text{z1020}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,1)=\frac{\sqrt{6}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6}+\frac{\sqrt{6}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z1021}}$ $$\begin{aligned}\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,2)=&-\frac{\sqrt{14}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{28}-\frac{\sqrt{14}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{28}+\frac{\sqrt{42}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{28}\\&+\frac{\sqrt{42}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{28}+\frac{\sqrt{42}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21}+\frac{\sqrt{42}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21}\end{aligned}$$

$\boxed{\text{z1022}}$ $$\begin{aligned}\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,2)=&-\frac{\sqrt{14}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{28}+\frac{\sqrt{42}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21}-\frac{\sqrt{42}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{28}\\&-\frac{\sqrt{14}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{28}-\frac{\sqrt{42}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{28}+\frac{\sqrt{42}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21}\end{aligned}$$

$\boxed{\text{z1023}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,2)=\frac{\sqrt{14}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14}+\frac{\sqrt{42}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21}+\frac{\sqrt{42}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21}+\frac{\sqrt{14}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14}$$

$\boxed{\text{z1024}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(T_u,a) = -\frac{\sqrt{6}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6}$

$\boxed{\text{z1025}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(T_u,a) = \frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6}$

$\boxed{\text{z1026}}$ $\mathbb{G}_{2,3}^{(1,0;c)}(T_u,a) = -\frac{\sqrt{2}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1027}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$

$\boxed{\text{z1028}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$

$\boxed{\text{z1029}}$ $\mathbb{G}_{2,3}^{(1,0;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$

$\boxed{\text{z1030}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$

$\boxed{\text{z1031}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$

$\boxed{\text{z1032}}$ $\mathbb{G}_{2,3}^{(1,1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$

- `A;A_001_1` : `'A'-'A'` bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `12a@12j`

$\boxed{\text{z22}}$ $\mathbb{Q}_{0}^{(c)}(A_g,a) = \mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{0}^{(b)}(A_g)$

$\boxed{\text{z23}}$ $\mathbb{Q}_{0}^{(c)}(A_g,b) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{5}$

$\boxed{\text{z24}}$ $\mathbb{Q}_{0}^{(c)}(A_g,c) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$

$\boxed{\text{z25}}$ $$\mathbb{Q}_4^{(c)}(A_g) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15}$$

$\boxed{\text{z26}}$ $$\mathbb{Q}_0^{(1,-1;c)}(A_g, a) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{5} + \frac{\sqrt{5}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{5}$$

$\boxed{\text{z27}}$ $$\mathbb{Q}_0^{(1,-1;c)}(A_g, b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z28}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_g, a) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15}$$

$\boxed{\text{z29}}$ $$\mathbb{Q}_4^{(1,-1;c)}(A_g, b) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z30}}$ $$\mathbb{Q}_0^{(1,0;c)}(A_g) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z31}}$ $$\mathbb{Q}_0^{(1,1;c)}(A_g, a) = \mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$$

$\boxed{\text{z32}}$ $$\mathbb{Q}_0^{(1,1;c)}(A_g, b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z33}}$ $$\mathbb{G}_3^{(c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z34}}$ $$\mathbb{G}_3^{(1,-1;c)}(A_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z35}}$ $$\mathbb{G}_3^{(1,-1;c)}(A_g, b) = \mathbb{M}_3^{(1,-1;a)}(A_g)\mathbb{T}_0^{(b)}(A_g)$$

$\boxed{\text{z36}}$ $$\mathbb{G}_3^{(1,-1;c)}(A_g, c) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z37}}$ $$\mathbb{G}_3^{(1,0;c)}(A_g, a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z38}}$ $$\mathbb{G}_3^{(1,0;c)}(A_g, b) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z39}}$ $$\mathbb{G}_3^{(1,0;c)}(A_g,c) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z112}}$ $$\mathbb{Q}_3^{(c)}(A_u,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z113}}$ $$\mathbb{Q}_3^{(c)}(A_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z114}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_u,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z115}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z116}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_u,c) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z117}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_u,d) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z118}}$ $$\mathbb{Q}_3^{(1,0;c)}(A_u,a) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z119}}$ $$\mathbb{Q}_3^{(1,0;c)}(A_u,b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z120}}$ $$\mathbb{G}_0^{(c)}(A_u,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z121}}$ $$\mathbb{G}_0^{(c)}(A_u,b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z122}}$ $$\mathbb{G}_0^{(1,-1;c)}(A_u,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z123}}$ $$\mathbb{G}_0^{(1,-1;c)}(A_u,b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z124}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_u,a) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z125}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_u,b) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z126}}$ $$\mathbb{G}_0^{(1,0;c)}(A_u) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z127}}$ $$\mathbb{G}_4^{(1,0;c)}(A_u) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z128}}$ $$\mathbb{G}_0^{(1,1;c)}(A_u,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z129}}$ $$\mathbb{G}_0^{(1,1;c)}(A_u,b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z130}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z131}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z132}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$\boxed{\text{z133}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$\boxed{\text{z134}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,c) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{7} + \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{7} + \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{7}$$

$\boxed{\text{z135}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,c) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{7} - \frac{\sqrt{21}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{7} + \frac{\sqrt{21}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14}$$

$\boxed{\text{z136}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,d) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z137}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,d) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$\boxed{\text{z138}}$ $$\mathbb{Q}_{4,1}^{(c)}(E_g) = \frac{\sqrt{21}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14} - \frac{\sqrt{21}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21} - \frac{\sqrt{21}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} - \frac{\sqrt{21}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21} + \frac{2\sqrt{21}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21}$$

$\boxed{\text{z139}}$ $$\mathbb{Q}_{4,2}^{(c)}(E_g) = -\frac{\sqrt{21}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} + \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{7} - \frac{\sqrt{21}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{7}$$

$\boxed{\text{z140}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$$

$\boxed{\text{z141}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$$

$\boxed{\text{z142}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, b) = \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{7} + \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{7} + \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{7}$$

$\boxed{\text{z143}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, b) = -\frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{7} - \frac{\sqrt{21}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{7} + \frac{\sqrt{21}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14}$$

$\boxed{\text{z144}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, c) = -\frac{\sqrt{2}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z145}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, c) = \frac{\sqrt{2}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z146}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, d) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g)}{28} + \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,3}^{(b)}(T_g)}{14}$$

$\boxed{\text{z147}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, d) = \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,1}^{(b)}(T_g)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,1}^{(b)}(T_g)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 1)\mathbb{M}_{1,2}^{(b)}(T_g)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,2}^{(b)}(T_g)}{84} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g, 2)\mathbb{M}_{1,3}^{(b)}(T_g)}{42}$$

$\boxed{\text{z361}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g, e) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z362}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g, e) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$\boxed{\text{z363}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g, a) = \frac{\sqrt{21}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14} - \frac{\sqrt{21}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21} - \frac{\sqrt{21}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} - \frac{\sqrt{21}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21} + \frac{2\sqrt{21}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21}$$

$\boxed{\text{z364}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g, a) = -\frac{\sqrt{21}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} + \frac{\sqrt{7}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{7} - \frac{\sqrt{21}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14} - \frac{\sqrt{7}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{7}$$

$\boxed{\text{z365}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(E_g,b) = -\frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{84} + \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{28} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{42}$$

$\boxed{\text{z366}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(E_g,b) = \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{14}$$

$\boxed{\text{z367}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{2}$$

$\boxed{\text{z368}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z369}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{0}^{(b)}(A_g)}{2}$$

$\boxed{\text{z370}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,b) = \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{0}^{(b)}(A_g)}{2}$$

$\boxed{\text{z371}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,c) = \frac{\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z372}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,c) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z373}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g,d) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{2} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$$

$\boxed{\text{z374}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g,d) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z375}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z376}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z377}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E_g,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z378}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g,b) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{2}$

$\boxed{\text{z379}}$ $\mathbb{Q}_{5,1}^{(c)}(E_u) = \frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{2}$

$\boxed{\text{z380}}$ $\mathbb{Q}_{5,2}^{(c)}(E_u) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$

$\boxed{\text{z381}}$ $\mathbb{Q}_{5,1}^{(1,-1;c)}(E_u) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{2}$

$\boxed{\text{z382}}$ $\mathbb{Q}_{5,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$

$\boxed{\text{z383}}$ $\mathbb{G}_{2,1}^{(c)}(E_u,a) = -\frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z384}}$ $\mathbb{G}_{2,2}^{(c)}(E_u,a) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z385}}$ $\mathbb{G}_{2,1}^{(c)}(E_u,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$

$\boxed{\text{z386}}$ $\mathbb{G}_{2,2}^{(c)}(E_u,b) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{2}$

$\boxed{\text{z387}}$ $\mathbb{G}_{2,1}^{(c)}(E_u,c) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$

$\boxed{\text{z388}}$ $\mathbb{G}_{2,2}^{(c)}(E_u,c) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{2}$

$\boxed{\text{z389}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,a) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z390}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,a) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z391}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,b) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{28} + \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{14}$$

$\boxed{\text{z392}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,b) = \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{84} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{42}$$

$\boxed{\text{z393}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,c) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{28} + \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{14}$$

$\boxed{\text{z394}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,c) = \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{84} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{42}$$

$\boxed{\text{z395}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,d) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z396}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,d) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{2}$$

$\boxed{\text{z397}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,e) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z398}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,e) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{2}$$

$\boxed{\text{z399}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,a) = -\frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{84} + \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{28} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{42}$$

$\boxed{\text{z400}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,a) = \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{14}$$

$\boxed{\text{z401}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(E_u,b) = -\frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{84} + \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{28} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{42}$$

$\boxed{\text{z402}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(E_u,b) = \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{14}$$

$\boxed{\text{z403}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E_u,a) = -\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z404}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E_u,a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z405}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E_u,b) = -\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6} - \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} + \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z406}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E_u,b) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{2} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{2}$$

$\boxed{\text{z407}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E_u,c) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{2} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{2}$$

$\boxed{\text{z408}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E_u,c) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z409}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E_u,d) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{2} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{2}$$

$\boxed{\text{z410}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E_u,d) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z411}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u,a) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z412}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u,a) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{2}$$

$\boxed{\text{z413}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z414}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u,b) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{2}$$

$\boxed{\text{z415}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z416}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z417}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z418}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z419}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z420}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z421}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,c) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{42} + \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{42} - \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} - \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14}$$

$\boxed{\text{z422}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,c) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{42} + \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{42} + \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14}$$

$\boxed{\text{z423}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,c) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21} + \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{21}$$

$\boxed{\text{z424}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,d) = -\frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z425}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,d) = \frac{\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z426}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,d) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z427}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,e) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z428}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,e) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z429}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,e) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z430}}$ $$\mathbb{Q}_{4,1}^{(c)}(T_g,1) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{4} - \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{4} - \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{12} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{12}$$

$\boxed{\text{z431}}$ $$\mathbb{Q}_{4,2}^{(c)}(T_g,1) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{4} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{12} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{4} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{12}$$

$\boxed{\text{z432}}$ $$\mathbb{Q}_{4,3}^{(c)}(T_g,1) = \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z433}}$ $$\mathbb{Q}_{4,1}^{(c)}(T_g,2) = -\frac{\sqrt{14}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{28} - \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{28} + \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{28} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{28} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21} + \frac{\sqrt{42}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21}$$

$\boxed{\text{z434}}$ $$\mathbb{Q}_{4,2}^{(c)}(T_g,2) = -\frac{\sqrt{14}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{28} + \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{28} - \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{28} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{28} + \frac{\sqrt{42}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21}$$

$\boxed{\text{z435}}$ $$\mathbb{Q}_{4,3}^{(c)}(T_g,2) = \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21} + \frac{\sqrt{14}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14}$$

$\boxed{\text{z436}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z437}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z438}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z439}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{42} + \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{42} - \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} - \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14}$$

$\boxed{\text{z440}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{42} + \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{42} + \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14}$$

$\boxed{\text{z441}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21} + \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{14} + \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{14} - \frac{\sqrt{42}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{21}$$

$\boxed{\text{z442}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,1}^{(b)}(E_g)}{12} + \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,2}^{(b)}(E_g)}{12} + \frac{\sqrt{10}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,1}^{(b)}(E_g)}{12} - \frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,2}^{(b)}(E_g)}{12}$$

$\boxed{\text{z443}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,c) = -\frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,1}^{(b)}(E_g)}{12} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,2}^{(b)}(E_g)}{12} + \frac{\sqrt{10}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,1}^{(b)}(E_g)}{12} + \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,2}^{(b)}(E_g)}{12}$$

$\boxed{\text{z444}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,c) = -\frac{\sqrt{2}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} - \frac{\sqrt{10}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,1}^{(b)}(E_g)}{6}$$

$\boxed{\text{z445}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{21}$$

$\boxed{\text{z446}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{21}$$

$\boxed{\text{z447}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{21}$$

$\boxed{\text{z448}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,e) = -\frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z449}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,e) = \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z450}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,e) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z451}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,f)=\frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z452}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,f)=\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z453}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,f)=\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z454}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,1a)=-\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{4}-\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{4}-\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{12}-\frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{12}$$

$\boxed{\text{z455}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,1a)=\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{4}-\frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{12}+\frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{4}-\frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{12}$$

$\boxed{\text{z456}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,1a)=\frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6}+\frac{\sqrt{6}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z457}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,1b)=-\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}-\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z458}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,1b)=\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,1}^{(b)}(E_g)}{2}-\frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z459}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,1b)=\frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z460}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,1c)=\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{4}-\frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{4}$$

$\boxed{\text{z461}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,1c)=-\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{4}+\frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{4}$$

$\boxed{\text{z462}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,1c)=\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{4}-\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{4}$$

$\boxed{\text{z463}}$ $$\begin{aligned}\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,2a)=&-\frac{\sqrt{14}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{28}-\frac{\sqrt{14}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{28}+\frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{28}\\&+\frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{28}+\frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21}+\frac{\sqrt{42}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21}\end{aligned}$$

$\boxed{\text{z464}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,2a) = -\frac{\sqrt{14}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{28} + \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{21} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{28} - \frac{\sqrt{14}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{28} - \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{28} + \frac{\sqrt{42}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21}$$

$\boxed{\text{z465}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,2a) = \frac{\sqrt{14}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{14} + \frac{\sqrt{42}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{21} + \frac{\sqrt{42}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{21} + \frac{\sqrt{14}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{14}$$

$\boxed{\text{z466}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,2b) = -\frac{\sqrt{30}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,1}^{(b)}(E_g)}{12} - \frac{\sqrt{10}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,2}^{(b)}(E_g)}{12} + \frac{\sqrt{2}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,1}^{(b)}(E_g)}{12} - \frac{\sqrt{6}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,2}^{(b)}(E_g)}{12}$$

$\boxed{\text{z467}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,2b) = \frac{\sqrt{30}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,1}^{(b)}(E_g)}{12} - \frac{\sqrt{10}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,2}^{(b)}(E_g)}{12} + \frac{\sqrt{2}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,1}^{(b)}(E_g)}{12} + \frac{\sqrt{6}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,2}^{(b)}(E_g)}{12}$$

$\boxed{\text{z468}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,2b) = \frac{\sqrt{10}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,2}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{2,1}^{(b)}(E_g)}{6}$$

$\boxed{\text{z469}}$ $$\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,2c) = -\frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{7}$$

$\boxed{\text{z470}}$ $$\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,2c) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{7}$$

$\boxed{\text{z471}}$ $$\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,2c) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{7}$$

$\boxed{\text{z472}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g,a) = -\frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z473}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g,a) = \frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} - \frac{\sqrt{2}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z474}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_g,a) = -\frac{\sqrt{2}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z475}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z476}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z477}}$ $\mathbb{Q}_{2,3}^{(1,0;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z478}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(T_g,c) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z479}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(T_g,c) = \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z480}}$ $\mathbb{Q}_{2,3}^{(1,0;c)}(T_g,c) = -\frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z481}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(T_g,d) = \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$

$\boxed{\text{z482}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(T_g,d) = -\frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$

$\boxed{\text{z483}}$ $\mathbb{Q}_{2,3}^{(1,0;c)}(T_g,d) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$

$\boxed{\text{z484}}$ $\mathbb{Q}_{4,1}^{(1,0;c)}(T_g,1) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z485}}$ $\mathbb{Q}_{4,2}^{(1,0;c)}(T_g,1) = \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z486}}$ $\mathbb{Q}_{4,3}^{(1,0;c)}(T_g,1) = \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z487}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$

$\boxed{\text{z488}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$

$\boxed{\text{z489}}$ $\mathbb{Q}_{2,3}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3}$

$\boxed{\text{z490}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(T_g,b) = -\frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z491}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(T_g,b) = \frac{\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{6}$$

$\boxed{\text{z492}}$ $$\mathbb{Q}_{2,3}^{(1,1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z493}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z494}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z495}}$ $$\mathbb{Q}_{2,3}^{(1,1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z496}}$ $$\mathbb{G}_{1,1}^{(c)}(T_g,a) = -\frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} + \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30}$$

$\boxed{\text{z497}}$ $$\mathbb{G}_{1,2}^{(c)}(T_g,a) = \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10} + \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30} - \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30}$$

$\boxed{\text{z498}}$ $$\mathbb{G}_{1,3}^{(c)}(T_g,a) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{15}$$

$\boxed{\text{z499}}$ $$\mathbb{G}_{1,1}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z500}}$ $$\mathbb{G}_{1,2}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z501}}$ $$\mathbb{G}_{1,3}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z502}}$ $$\mathbb{G}_{1,1}^{(c)}(T_g,c) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z503}}$ $$\mathbb{G}_{1,2}^{(c)}(T_g,c) = -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z504}}$ $$\mathbb{G}_{1,3}^{(c)}(T_g,c) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$\boxed{\text{z505}}$ $$\mathbb{G}_{1,1}^{(c)}(T_g,d) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z506}}$ $$\mathbb{G}_{1,2}^{(c)}(T_g,d) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z507}}$ $$\mathbb{G}_{1,3}^{(c)}(T_g,d) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z508}}$ $$\mathbb{G}_{3,1}^{(c)}(T_g,1) = \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{20} - \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{20} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{60} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{60} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15}$$

$\boxed{\text{z509}}$ $$\mathbb{G}_{3,2}^{(c)}(T_g,1) = -\frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{20} + \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{60} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{20} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{60} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z510}}$ $$\mathbb{G}_{3,3}^{(c)}(T_g,1) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30}$$

$\boxed{\text{z511}}$ $$\mathbb{G}_{3,1}^{(c)}(T_g,2) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{12} - \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{12} + \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{4} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{4}$$

$\boxed{\text{z512}}$ $$\mathbb{G}_{3,2}^{(c)}(T_g,2) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{12} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{4} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{12} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{4}$$

$\boxed{\text{z513}}$ $$\mathbb{G}_{3,3}^{(c)}(T_g,2) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6}$$

$\boxed{\text{z514}}$ $$\begin{aligned}\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,a) = &-\frac{\sqrt{10}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} + \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30} \\ &- \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30}\end{aligned}$$

$\boxed{\text{z515}}$ $$\begin{aligned}\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,a) = &\frac{\sqrt{10}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10} + \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30} \\ &- \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30}\end{aligned}$$

$\boxed{\text{z516}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,a) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{30} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{15}$$

$\boxed{\text{z517}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z518}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z519}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$\boxed{\text{z520}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,c) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z521}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,c) = \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z522}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,c) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z702}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z703}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,d) = -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z704}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,d) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$\boxed{\text{z705}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g,e) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z706}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g,e) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z707}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g,e) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z708}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1a) = \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{20} - \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{20} - \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{60} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{60} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15}$$

$\boxed{\text{z709}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1a) = -\frac{\sqrt{10}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{20} + \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{60} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{20} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{60} - \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z710}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1a) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{30} + \frac{\sqrt{30}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{30}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{30}$$

$\boxed{\text{z711}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1b) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z712}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1b) = \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z713}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1b) = \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z714}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1c) = -\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{12}$$

$\boxed{\text{z715}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1c) = \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{12}$$

$\boxed{\text{z716}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1c) = -\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{12}$$

$\boxed{\text{z717}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2a) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{12} - \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{12} + \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{4} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{4}$$

$\boxed{\text{z718}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2a) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{12} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{4} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{12} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{4}$$

$\boxed{\text{z719}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2a) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6}$$

$\boxed{\text{z784}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2b) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z785}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2b) = \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z786}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2b)=\frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z787}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2c)=\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{12}+\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{12}+\frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{12}+\frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z788}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2c)=\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,3}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,3}^{(b)}(T_g)}{12}+\frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{12}+\frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{12}+\frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z789}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2c)=\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,2}^{(b)}(T_g)}{12}+\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,2}^{(b)}(T_g)}{12}+\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{M}_{1,1}^{(b)}(T_g)}{12}-\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{M}_{1,1}^{(b)}(T_g)}{12}+\frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3}$$

$\boxed{\text{z790}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_g,a)=\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z791}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_g,a)=\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z792}}$ $$\mathbb{G}_{1,3}^{(1,0;c)}(T_g,a)=\frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z793}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_g,b)=-\frac{\sqrt{30}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{30}+\frac{\sqrt{10}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10}+\frac{\sqrt{10}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{10}+\frac{\sqrt{10}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10}$$

$\boxed{\text{z794}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_g,b)=\frac{\sqrt{10}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{10}-\frac{\sqrt{30}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{30}-\frac{\sqrt{10}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10}+\frac{\sqrt{10}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10}$$

$\boxed{\text{z795}}$ $$\mathbb{G}_{1,3}^{(1,0;c)}(T_g,b)=\frac{\sqrt{10}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{10}+\frac{\sqrt{10}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{10}+\frac{\sqrt{30}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{15}$$

$\boxed{\text{z796}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_g,c)=-\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{30}+\frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{10}+\frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{10}+\frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{10}$$

$\boxed{\text{z797}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_g,c)=-\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{30}+\frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{10}-\frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{10}+\frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{10}$$

$\boxed{\text{z798}}$ $$\mathbb{G}_{1,3}^{(1,0;c)}(T_g,c)=\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{15}+\frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{10}+\frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{10}$$

$\boxed{\text{z799}}$ $$\mathbb{G}_{3,1}^{(1,0;c)}(T_g,1a)=-\frac{\sqrt{5}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10}+\frac{\sqrt{15}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10}-\frac{\sqrt{15}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15}-\frac{\sqrt{15}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15}$$

$\boxed{\text{z800}}$ $$\mathbb{G}_{3,2}^{(1,0;c)}(T_g,1a) = -\frac{\sqrt{15}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{15} - \frac{\sqrt{5}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{10} - \frac{\sqrt{15}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{10} - \frac{\sqrt{15}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z801}}$ $$\mathbb{G}_{3,3}^{(1,0;c)}(T_g,1a) = -\frac{\sqrt{15}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{15} + \frac{\sqrt{5}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{5}$$

$\boxed{\text{z802}}$ $$\mathbb{G}_{3,1}^{(1,0;c)}(T_g,1b) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{10} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{15}$$

$\boxed{\text{z803}}$ $$\mathbb{G}_{3,2}^{(1,0;c)}(T_g,1b) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z804}}$ $$\mathbb{G}_{3,3}^{(1,0;c)}(T_g,1b) = \frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{5} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{15}$$

$\boxed{\text{z805}}$ $$\mathbb{G}_{3,1}^{(1,0;c)}(T_g,2a) = -\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} + \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3} - \frac{\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z806}}$ $$\mathbb{G}_{3,2}^{(1,0;c)}(T_g,2a) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,3}^{(b)}(T_g)}{3} + \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6} + \frac{\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z807}}$ $$\mathbb{G}_{3,3}^{(1,0;c)}(T_g,2a) = \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(T_g)}{3} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(T_g)}{3} + \frac{\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$$

$\boxed{\text{z808}}$ $$\mathbb{G}_{3,1}^{(1,0;c)}(T_g,2b) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3} + \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{3}$$

$\boxed{\text{z809}}$ $$\mathbb{G}_{3,2}^{(1,0;c)}(T_g,2b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} + \frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z810}}$ $$\mathbb{G}_{3,3}^{(1,0;c)}(T_g,2b) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{3}$$

$\boxed{\text{z811}}$ $$\mathbb{G}_{1,1}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z812}}$ $$\mathbb{G}_{1,2}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z813}}$ $$\mathbb{G}_{1,3}^{(1,1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z814}}$ $$\mathbb{G}_{1,1}^{(1,1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z815}}$ $$\mathbb{G}_{1,2}^{(1,1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z816}}$ $$\mathbb{G}_{1,3}^{(1,1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{2,1}^{(b)}(E_g)}{3}$$

$\boxed{\text{z817}}$ $$\mathbb{G}_{1,1}^{(1,1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6}$$

$\boxed{\text{z818}}$ $$\mathbb{G}_{1,2}^{(1,1;c)}(T_g,c) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,3}^{(b)}(T_g)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z819}}$ $$\mathbb{G}_{1,3}^{(1,1;c)}(T_g,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{M}_{1,2}^{(b)}(T_g)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{M}_{1,1}^{(b)}(T_g)}{6}$$

$\boxed{\text{z1033}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1034}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1035}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1036}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u,b) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1037}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u,b) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{10} - \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1038}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u,b) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{15} + \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1039}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u,c) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{10} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{15}$$

$\boxed{\text{z1040}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u,c) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{15}$$

$\boxed{\text{z1041}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u,c) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{5} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{15}$$

$\boxed{\text{z1042}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u,d) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1043}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u,d) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1044}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u,d) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1045}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u,e) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1046}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u,e) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1047}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u,e) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1048}}$ $$\mathbb{Q}_{3,1}^{(c)}(T_u,1a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z1049}}$ $$\mathbb{Q}_{3,2}^{(c)}(T_u,1a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z1050}}$ $$\mathbb{Q}_{3,3}^{(c)}(T_u,1a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z1051}}$ $$\mathbb{Q}_{3,1}^{(c)}(T_u,1b) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1052}}$ $$\mathbb{Q}_{3,2}^{(c)}(T_u,1b) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1053}}$ $$\mathbb{Q}_{3,3}^{(c)}(T_u,1b) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1054}}$ $$\mathbb{Q}_{3,1}^{(c)}(T_u,1c) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{30} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{10}$$

$\boxed{\text{z1055}}$ $$\mathbb{Q}^{(c)}_{3,2}(T_u,1c) = -\frac{\sqrt{30}\mathbb{Q}^{(a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{30} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{3,3}(T_u,1)}{10} - \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{10}$$

$\boxed{\text{z1056}}$ $$\mathbb{Q}^{(c)}_{3,3}(T_u,1c) = \frac{\sqrt{30}\mathbb{Q}^{(a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{3,3}(T_u,1)}{15} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{10}$$

$\boxed{\text{z1057}}$ $$\mathbb{Q}^{(c)}_{3,1}(T_u,2a) = -\frac{\sqrt{3}\mathbb{Q}^{(a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{1,1}(T_u)}{6} - \frac{\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{1,1}(T_u)}{6} - \frac{\mathbb{Q}^{(a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{1,3}(T_u)}{3} + \frac{\mathbb{Q}^{(a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{1,2}(T_u)}{3}$$

$\boxed{\text{z1058}}$ $$\mathbb{Q}^{(c)}_{3,2}(T_u,2a) = \frac{\sqrt{3}\mathbb{Q}^{(a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{1,2}(T_u)}{6} + \frac{\mathbb{Q}^{(a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{1,3}(T_u)}{3} - \frac{\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{1,2}(T_u)}{6} - \frac{\mathbb{Q}^{(a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{1,1}(T_u)}{3}$$

$\boxed{\text{z1059}}$ $$\mathbb{Q}^{(c)}_{3,3}(T_u,2a) = -\frac{\mathbb{Q}^{(a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{1,2}(T_u)}{3} + \frac{\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{1,3}(T_u)}{3} + \frac{\mathbb{Q}^{(a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{1,1}(T_u)}{3}$$

$\boxed{\text{z1060}}$ $$\mathbb{Q}^{(c)}_{3,1}(T_u,2b) = \frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{3} + \frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{9} - \frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{3,3}(T_u,1)}{18} + \frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{18}$$

$\boxed{\text{z1061}}$ $$\mathbb{Q}^{(c)}_{3,2}(T_u,2b) = -\frac{\sqrt{2}\mathbb{Q}^{(a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{3} + \frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{3,3}(T_u,1)}{18} + \frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{9} - \frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{18}$$

$\boxed{\text{z1062}}$ $$\mathbb{Q}^{(c)}_{3,3}(T_u,2b) = -\frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{18} - \frac{2\sqrt{6}\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{3,3}(T_u,1)}{9} + \frac{\sqrt{6}\mathbb{Q}^{(a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{18}$$

$\boxed{\text{z1063}}$ $$\mathbb{Q}^{(c)}_{5,1}(T_u,3) = \frac{\mathbb{Q}^{(a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{6} + \frac{\sqrt{3}\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{18} + \frac{2\sqrt{3}\mathbb{Q}^{(a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{3,3}(T_u,1)}{9} - \frac{2\sqrt{3}\mathbb{Q}^{(a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{9}$$

$\boxed{\text{z1064}}$ $$\mathbb{Q}^{(c)}_{5,2}(T_u,3) = -\frac{\mathbb{Q}^{(a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{6} - \frac{2\sqrt{3}\mathbb{Q}^{(a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{3,3}(T_u,1)}{9} + \frac{\sqrt{3}\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{18} + \frac{2\sqrt{3}\mathbb{Q}^{(a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{9}$$

$\boxed{\text{z1065}}$ $$\mathbb{Q}^{(c)}_{5,3}(T_u,3) = \frac{2\sqrt{3}\mathbb{Q}^{(a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{3,2}(T_u,1)}{9} - \frac{\sqrt{3}\mathbb{Q}^{(a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{3,3}(T_u,1)}{9} - \frac{2\sqrt{3}\mathbb{Q}^{(a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{3,1}(T_u,1)}{9}$$

$\boxed{\text{z1066}}$ $$\mathbb{Q}^{(1,-1;c)}_{1,1}(T_u,a) = -\frac{\sqrt{30}\mathbb{Q}^{(1,-1;a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{1,1}(T_u)}{30} + \frac{\sqrt{10}\mathbb{Q}^{(1,-1;a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{1,1}(T_u)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(1,-1;a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{1,3}(T_u)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(1,-1;a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{1,2}(T_u)}{10}$$

$\boxed{\text{z1067}}$ $$\mathbb{Q}^{(1,-1;c)}_{1,2}(T_u,a) = -\frac{\sqrt{30}\mathbb{Q}^{(1,-1;a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{1,2}(T_u)}{30} + \frac{\sqrt{10}\mathbb{Q}^{(1,-1;a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{1,3}(T_u)}{10} - \frac{\sqrt{10}\mathbb{Q}^{(1,-1;a)}_{2,2}(E_g)\mathbb{Q}^{(b)}_{1,2}(T_u)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(1,-1;a)}_{2,3}(T_g)\mathbb{Q}^{(b)}_{1,1}(T_u)}{10}$$

$\boxed{\text{z1068}}$ $$\mathbb{Q}^{(1,-1;c)}_{1,3}(T_u,a) = \frac{\sqrt{30}\mathbb{Q}^{(1,-1;a)}_{2,1}(E_g)\mathbb{Q}^{(b)}_{1,3}(T_u)}{15} + \frac{\sqrt{10}\mathbb{Q}^{(1,-1;a)}_{2,1}(T_g)\mathbb{Q}^{(b)}_{1,2}(T_u)}{10} + \frac{\sqrt{10}\mathbb{Q}^{(1,-1;a)}_{2,2}(T_g)\mathbb{Q}^{(b)}_{1,1}(T_u)}{10}$$

$\boxed{\text{z1069}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u,b) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{10} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{15}$$

$\boxed{\text{z1070}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u,b) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{15}$$

$\boxed{\text{z1071}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{5} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{15}$$

$\boxed{\text{z1072}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u,c) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1073}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u,c) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1074}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1075}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u,d) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1076}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u,d) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1077}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u,d) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1078}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1a) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1079}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1a) = -\frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1080}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1a) = \frac{\sqrt{5}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1081}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1b) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{30} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{10}$$

$\boxed{\text{z1082}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1b) = -\frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{30} + \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{10} - \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{10}$$

$\boxed{\text{z1083}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1b) = \frac{\sqrt{30}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{15} + \frac{\sqrt{10}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{10} + \frac{\sqrt{10}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{10}$$

$\boxed{\text{z1084}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1c) = -\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{12}$$

$\boxed{\text{z1085}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1c) = \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{12}$$

$\boxed{\text{z1086}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1c) = -\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{12}$$

$\boxed{\text{z1087}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1d) = -\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{12}$$

$\boxed{\text{z1088}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1d) = \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{12}$$

$\boxed{\text{z1089}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1d) = -\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{12}$$

$\boxed{\text{z1090}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2a) = -\frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1091}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1092}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2a) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1093}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3} + \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{9} - \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{18} + \frac{\sqrt{6}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{18}$$

$\boxed{\text{z1094}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2b) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3} + \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{18} + \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{9} - \frac{\sqrt{6}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{18}$$

$\boxed{\text{z1095}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2b) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{18} - \frac{2\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{9} + \frac{\sqrt{6}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{18}$$

$\boxed{\text{z1096}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2c) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z1097}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2c) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z1098}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2c) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{12} + \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z1099}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2d) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z1100}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2d) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z1101}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2d) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{12} + \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z1102}}$ $$\mathbb{Q}_{5,1}^{(1,-1;c)}(T_u,3) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{18} + \frac{2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{9} - \frac{2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{9}$$

$\boxed{\text{z1103}}$ $$\mathbb{Q}_{5,2}^{(1,-1;c)}(T_u,3) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} - \frac{2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{9} + \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{18} + \frac{2\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{9}$$

$\boxed{\text{z1104}}$ $$\mathbb{Q}_{5,3}^{(1,-1;c)}(T_u,3) = \frac{2\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{9} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{9} - \frac{2\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{9}$$

$\boxed{\text{z1105}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1106}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,a) = -\frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1107}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1108}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10}$$

$\boxed{\text{z1109}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{10} - \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10}$$

$\boxed{\text{z1110}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,b) = \frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{15} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10}$$

$\boxed{\text{z1111}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,c) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10}$$

$\boxed{\text{z1112}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,c) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{10} - \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10}$$

$\boxed{\text{z1113}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,c) = \frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{15} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10}$$

$\boxed{\text{z1114}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,1a) = -\frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z1115}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,1a) = \frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z1116}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,1a) = -\frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} + \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z1117}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,1b) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{15}$$

$\boxed{\text{z1118}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,1b) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{15}$$

$\boxed{\text{z1119}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,1b) = \frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{5} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{15}$$

$\boxed{\text{z1120}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,1c) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{15}$$

$\boxed{\text{z1121}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,1c) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{15}$$

$\boxed{\text{z1122}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,1c) = \frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{5} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{15}$$

$\boxed{\text{z1123}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2a) = -\frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z1124}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2a) = -\frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z1125}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2a) = -\frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{6} - \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{6}$$

$\boxed{\text{z1126}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2b) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3} + \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z1127}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2b) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z1128}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2b) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{3}$$

$\boxed{\text{z1129}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2c) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3} + \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z1130}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2c) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z1131}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2c) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{3}$$

$\boxed{\text{z1132}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1133}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1134}}$ $$\mathbb{Q}_{1,3}^{(1,1;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1135}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1136}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(T_u,b) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1137}}$ $$\mathbb{Q}_{1,3}^{(1,1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1138}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(T_u,c) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1139}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(T_u,c) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1140}}$ $$\mathbb{Q}_{1,3}^{(1,1;c)}(T_u,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1141}}$ $$\mathbb{Q}_{3,1}^{(1,1;c)}(T_u,1) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{3,1}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z1142}}$ $$\mathbb{Q}_{3,2}^{(1,1;c)}(T_u,1) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{3,2}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z1143}}$ $$\mathbb{Q}_{3,3}^{(1,1;c)}(T_u,1) = \frac{\sqrt{3}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{3,3}^{(b)}(T_u,1)}{3}$$

$\boxed{\text{z1144}}$ $$\mathbb{G}_{2,1}^{(c)}(T_u,a) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1145}}$ $$\mathbb{G}_{2,2}^{(c)}(T_u,a) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1146}}$ $$\mathbb{G}_{2,3}^{(c)}(T_u,a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1147}}$ $$\mathbb{G}_{2,1}^{(c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1148}}$ $$\mathbb{G}_{2,2}^{(c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1149}}$ $$\mathbb{G}_{2,3}^{(c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1150}}$ $$\mathbb{G}_{2,1}^{(c)}(T_u,c) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1151}}$ $$\mathbb{G}_{2,2}^{(c)}(T_u,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1152}}$ $$\mathbb{G}_{2,3}^{(c)}(T_u,c) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1153}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1154}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{6}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1155}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1156}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,b) = -\frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{21}$$

$\boxed{\text{z1157}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,b) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{21}$$

$\boxed{\text{z1158}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,b) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{21}$$

$\boxed{\text{z1159}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,c) = -\frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{21}$$

$\boxed{\text{z1160}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,c) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{21}$$

$\boxed{\text{z1161}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,c) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{21}$$

$\boxed{\text{z1162}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,d)=\frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1163}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,d)=\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1164}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,d)=\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1165}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,e)=\frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1166}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,e)=\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1167}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,e)=\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}+\frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$$

$\boxed{\text{z1168}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,1a)=\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{12}-\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{4}-\frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{12}-\frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{4}$$

$\boxed{\text{z1169}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,1a)=-\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{12}-\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{4}+\frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{12}-\frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{4}$$

$\boxed{\text{z1170}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,1a)=\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{12}-\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{4}-\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{12}-\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{4}$$

$\boxed{\text{z1171}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,1b)=\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{12}-\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{4}-\frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{12}-\frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{4}$$

$\boxed{\text{z1172}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,1b)=-\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{12}-\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{4}+\frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{12}-\frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{4}$$

$\boxed{\text{z1173}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,1b)=\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{12}-\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{4}-\frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{12}-\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{4}$$

$\boxed{\text{z1174}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,2a)=-\frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{84}-\frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{28}-\frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{84}$$
$$+\frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{28}+\frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{7}$$

$\boxed{\text{z1175}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,2a) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{7}$$

$\boxed{\text{z1176}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,2a) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{7}$$

$\boxed{\text{z1177}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,2b) = -\frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{7}$$

$\boxed{\text{z1178}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,2b) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{7}$$

$\boxed{\text{z1179}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,2b) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{7}$$

$\boxed{\text{z1180}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1181}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1182}}$ $$\mathbb{G}_{2,3}^{(1,0;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1183}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1184}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$$

$\boxed{\text{z1185}}$ $\mathbb{G}_{2,3}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{3} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z1186}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(T_u,c) = \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z1187}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(T_u,c) = -\frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z1188}}$ $\mathbb{G}_{2,3}^{(1,0;c)}(T_u,c) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{3} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z1189}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z1190}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z1191}}$ $\mathbb{G}_{2,3}^{(1,1;c)}(T_u,a) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,a)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,a)}{6}$

$\boxed{\text{z1192}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z1193}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

$\boxed{\text{z1194}}$ $\mathbb{G}_{2,3}^{(1,1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u,b)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u,b)}{6}$

- `A;A_002_1` : `'A'-'A'` bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `3a@1a`

$\boxed{\text{z40}}$ $\mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z148}}$ $\mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z149}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z720}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z820}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z821}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z1195}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1196}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1197}}$ $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1198}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1199}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1200}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

- `A;A_002_1` : `'A'-'A'` bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3a@1a`

$\boxed{\text{z41}}$ $\mathbb{Q}_{0}^{(c)}(A_g) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z42}}$ $\mathbb{Q}_{0}^{(1,0;c)}(A_g) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z43}}$ $\mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z150}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z151}}$ $\mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z152}}$ $\mathbb{G}_0^{(1,1;c)}(A_u) = \mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z153}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z154}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z155}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z523}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z524}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(E_g) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z525}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z526}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2}$

$\boxed{\text{z527}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2}$

$\boxed{\text{z528}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z529}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z530}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z531}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z532}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z533}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z534}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z535}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g) = \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z536}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g) = -\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z537}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z538}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z539}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z540}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z541}}$ $$\mathbb{G}_{1,1}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z542}}$ $$\mathbb{G}_{1,2}^{(c)}(T_g) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z543}}$ $$\mathbb{G}_{1,3}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z544}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10}$$

$\boxed{\text{z545}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} - \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z546}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g) = \frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z547}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{15}$$

$\boxed{\text{z548}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z549}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z721}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z722}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z723}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z822}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z823}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_g) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z824}}$ $$\mathbb{G}_{1,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z825}}$ $$\mathbb{G}_{1,1}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z826}}$ $$\mathbb{G}_{1,2}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z827}}$ $\mathbb{G}_{1,3}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_0^{(1,1;a)}(A_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z1201}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z1202}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z1203}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z1204}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1205}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1206}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1207}}$ $\mathbb{Q}_{3,1}^{(c)}(T_u, 2) = -\frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1208}}$ $\mathbb{Q}_{3,2}^{(c)}(T_u, 2) = \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1209}}$ $\mathbb{Q}_{3,3}^{(c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1210}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1211}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1212}}$ $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1213}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1214}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1215}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1216}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1217}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1218}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1219}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1220}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1221}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1222}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1223}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2) = \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1224}}$ $\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1225}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1226}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1227}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

- `A;A_002_1` : 'A'-'A' bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3a@1a`

$\boxed{\text{z44}}$ $\mathbb{Q}_0^{(c)}(A_g,a) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z45}}$ $\mathbb{Q}_0^{(c)}(A_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z46}}$ $\mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z47}}$ $\mathbb{Q}_0^{(1,1;c)}(A_g) = \mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z48}}$ $\mathbb{G}_3^{(c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z49}}$ $\mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z156}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_u) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z157}}$ $\mathbb{Q}_3^{(1,0;c)}(A_u) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z158}}$ $\mathbb{G}_0^{(c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z159}}$ $\mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z160}}$ $\mathbb{G}_4^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z161}}$ $\mathbb{G}_0^{(1,1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z162}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z163}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z164}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$

$\boxed{\text{z165}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$

$\boxed{\text{z166}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z167}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z550}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$

$\boxed{\text{z551}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$

$\boxed{\text{z552}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z553}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z554}}$ $\mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z555}}$ $\mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z556}}$ $\mathbb{G}_{2,1}^{(c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z557}}$ $\mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z558}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,a) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{14}$

$\boxed{\text{z559}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,a) = \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{42}$$

$\boxed{\text{z560}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z561}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z562}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} + \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{42}$$

$\boxed{\text{z563}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{14}$$

$\boxed{\text{z564}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E_u) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z565}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E_u) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z566}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z567}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z568}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z569}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z570}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$$

$\boxed{\text{z571}}$ $\mathbb{Q}_{2,1}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z572}}$ $\mathbb{Q}_{2,2}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z573}}$ $\mathbb{Q}_{2,3}^{(c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z574}}$ $\mathbb{Q}_{4,1}^{(c)}(T_g,1) = -\frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z575}}$ $\mathbb{Q}_{4,2}^{(c)}(T_g,1) = \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z576}}$ $\mathbb{Q}_{4,3}^{(c)}(T_g,1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z724}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z725}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z726}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z727}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z728}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z729}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z828}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,1) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z829}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,1) = \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z830}}$ $\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z831}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z832}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z833}}$ $\mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z834}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z835}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z836}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z837}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z838}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z839}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1228}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1229}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1230}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1231}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

z1232 $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

z1233 $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

z1234 $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1) = -\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

z1235 $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1) = \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

z1236 $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1) = -\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

z1237 $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

z1238 $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

z1239 $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

z1240 $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_u) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10}$$

z1241 $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} - \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

z1242 $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u) = \frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

z1243 $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{15}$$

z1244 $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

z1245 $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,1) = \frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1246}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1247}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1248}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1249}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1250}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1251}}$ $$\mathbb{Q}_{1,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1252}}$ $$\mathbb{G}_{2,1}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1253}}$ $$\mathbb{G}_{2,2}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1254}}$ $$\mathbb{G}_{2,3}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1255}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1256}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1257}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1258}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1259}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1260}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1261}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,1) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1262}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,1) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1263}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,1) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1264}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{7}$$

$\boxed{\text{z1265}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{7}$$

$\boxed{\text{z1266}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{7}$$

$\boxed{\text{z1267}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(T_u) = \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1268}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(T_u) = -\frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1269}}$ $$\mathbb{G}_{2,3}^{(1,0;c)}(T_u) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1270}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1271}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1272}}$ $$\mathbb{G}_{2,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

- `A;A_002_2` : 'A'-'A' bond-cluster

* bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
* ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
* wyckoff: `3b@3c`

$\boxed{\text{z50}}$ $\mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z168}}$ $\mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z169}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z730}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z840}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z841}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z1273}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1274}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1275}}$ $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1276}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1277}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1278}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

- `A;A_002_2` : '`A`'-'`A`' bond-cluster

* bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
* ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
* wyckoff: `3b@3c`

$\boxed{\text{z51}}$ $$\mathbb{Q}_0^{(c)}(A_g) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z52}}$ $$\mathbb{Q}_0^{(1,0;c)}(A_g) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z53}}$ $$\mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z170}}$ $$\mathbb{Q}_3^{(1,-1;c)}(A_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z171}}$ $$\mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z172}}$ $$\mathbb{G}_0^{(1,1;c)}(A_u) = \mathbb{G}_0^{(1,1;a)}(A_u)\mathbb{Q}_0^{(b)}(A_g)$$

$\boxed{\text{z173}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z174}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z175}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z577}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z578}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z579}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z580}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$\boxed{\text{z581}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$$

$\boxed{\text{z582}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z583}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z584}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z585}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z586}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z587}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z588}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z589}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g) = \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z590}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g) = -\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z591}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z592}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z593}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z594}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z595}}$ $$\mathbb{G}_{1,1}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z596}}$ $$\mathbb{G}_{1,2}^{(c)}(T_g) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z597}}$ $$\mathbb{G}_{1,3}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z598}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10}$$

$\boxed{\text{z599}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} - \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z600}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g) = \frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z601}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{15}$$

$\boxed{\text{z602}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z603}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z731}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z732}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z733}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z842}}$ $$\mathbb{G}_{1,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z843}}$ $$\mathbb{G}_{1,2}^{(1,0;c)}(T_g) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z844}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z845}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$

$\boxed{\text{z846}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$

$\boxed{\text{z847}}$ $\mathbb{G}_{1,3}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z1279}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1280}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1281}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1282}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1283}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1284}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u, b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1285}}$ $\mathbb{Q}_{3,1}^{(c)}(T_u, 2) = -\frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1286}}$ $\mathbb{Q}_{3,2}^{(c)}(T_u, 2) = \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1287}}$ $\mathbb{Q}_{3,3}^{(c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1288}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1289}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1290}}$ $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1291}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u, 2) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1292}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u, 2) = -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1293}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1294}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1295}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1296}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u, a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1297}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1298}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u, b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1299}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u, b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1300}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(T_u, 2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1301}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(T_u, 2) = \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1302}}$ $\mathbb{Q}_{3,3}^{(1,0;c)}(T_u, 2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1303}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z1304}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

$\boxed{\text{z1305}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_0^{(b)}(A_g)}{3}$

- `A;A_002_2` : `'A'-'A'` bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3b@3c`

$\boxed{\text{z54}}$ $\mathbb{Q}_0^{(c)}(A_g,a) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z55}}$ $\mathbb{Q}_0^{(c)}(A_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z56}}$ $\mathbb{Q}_0^{(1,-1;c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z57}}$ $\mathbb{Q}_0^{(1,1;c)}(A_g) = \mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z58}}$ $\mathbb{G}_3^{(c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z59}}$ $\mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z176}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_u) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z177}}$ $\mathbb{Q}_3^{(1,0;c)}(A_u) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z178}}$ $\mathbb{G}_0^{(c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z179}}$ $$\mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z180}}$ $$\mathbb{G}_4^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z181}}$ $$\mathbb{G}_0^{(1,1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z182}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z183}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z184}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$\boxed{\text{z185}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$\boxed{\text{z186}}$ $$\mathbb{Q}_{2,1}^{(c)}(E_g,c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z187}}$ $$\mathbb{Q}_{2,2}^{(c)}(E_g,c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z604}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$\boxed{\text{z605}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$$

$\boxed{\text{z606}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z607}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z608}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z609}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z610}}$ $$\mathbb{G}_{2,1}^{(c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z611}}$ $$\mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z612}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,a) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{14}$$

$\boxed{\text{z613}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,a) = \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{42}$$

$\boxed{\text{z614}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z615}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z616}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} + \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{42}$$

$\boxed{\text{z617}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{14}$$

$\boxed{\text{z618}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(E_u) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z619}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(E_u) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z620}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z621}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z622}}$ $\mathbb{Q}_{2,1}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z623}}$ $\mathbb{Q}_{2,2}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z624}}$ $\mathbb{Q}_{2,3}^{(c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z625}}$ $\mathbb{Q}_{2,1}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z626}}$ $\mathbb{Q}_{2,2}^{(c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z627}}$ $\mathbb{Q}_{2,3}^{(c)}(T_g, b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z628}}$ $\mathbb{Q}_{4,1}^{(c)}(T_g, 1) = -\frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z629}}$ $\mathbb{Q}_{4,2}^{(c)}(T_g, 1) = \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z630}}$ $\mathbb{Q}_{4,3}^{(c)}(T_g, 1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z734}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z735}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z736}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g, a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z737}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g, b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z738}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z739}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z848}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,1) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z849}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,1) = \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z850}}$ $\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z851}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z852}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z853}}$ $\mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z854}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z855}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z856}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z857}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z858}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z859}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1306}}$ $$\mathbb{Q}_{1,1}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1307}}$ $$\mathbb{Q}_{1,2}^{(c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1308}}$ $$\mathbb{Q}_{1,3}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1309}}$ $$\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1310}}$ $$\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1311}}$ $$\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1312}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1) = -\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12}$$

$\boxed{\text{z1313}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1) = \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$\boxed{\text{z1314}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1) = -\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$$

$\boxed{\text{z1315}}$ $$\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1316}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1317}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1318}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_u) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1319}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} - \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1320}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u) = \frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1321}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1322}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1323}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,1) = \frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1324}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1325}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1326}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1327}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1328}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1329}}$ $$\mathbb{Q}_{1,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1330}}$ $$\mathbb{G}_{2,1}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1331}}$ $$\mathbb{G}_{2,2}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1332}}$ $$\mathbb{G}_{2,3}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1333}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1334}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1335}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1336}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1337}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1338}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1339}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,1) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1340}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,1) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1341}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,1) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1342}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{7}$$

$\boxed{\text{z1343}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{7}$$

$\boxed{\text{z1344}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{7}$$

$\boxed{\text{z1345}}$ $$\mathbb{G}_{2,1}^{(1,0;c)}(T_u) = \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1346}}$ $$\mathbb{G}_{2,2}^{(1,0;c)}(T_u) = -\frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1347}}$ $$\mathbb{G}_{2,3}^{(1,0;c)}(T_u) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1348}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1349}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1350}}$ $\mathbb{G}_{2,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

- `A;A_002_3` : `'A'-'A'` bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$
  * wyckoff: `3c@3c`

$\boxed{\text{z60}}$ $\mathbb{Q}_0^{(c)}(A_g) = \mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_0^{(b)}(A_g)$

$\boxed{\text{z188}}$ $\mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z189}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z740}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z860}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z861}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z1351}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1352}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1353}}$ $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1354}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1355}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1356}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

- `A;A_002_3` : `'A'-'A'` bond-cluster
  * bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3c@3c`

$\boxed{\text{z61}}$ $\mathbb{Q}_{0}^{(c)}(A_g) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z62}}$ $\mathbb{Q}_{0}^{(1,0;c)}(A_g) = \frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z63}}$ $\mathbb{G}_{3}^{(1,-1;c)}(A_g) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z190}}$ $\mathbb{Q}_{3}^{(1,-1;c)}(A_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z191}}$ $\mathbb{G}_{0}^{(1,-1;c)}(A_u) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z192}}$ $\mathbb{G}_{0}^{(1,1;c)}(A_u) = \mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{0}^{(b)}(A_g)$

$\boxed{\text{z193}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z194}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z195}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z631}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z632}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(E_g) = -\frac{\sqrt{3}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z633}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(E_g) = \frac{\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z634}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$$

$\boxed{\text{z635}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u, a) = \frac{\sqrt{2}\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$$

$\boxed{\text{z636}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u, b) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z637}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u, b) = -\frac{\mathbb{G}_{2,1}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(E_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z638}}$ $$\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z639}}$ $$\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\sqrt{2}\mathbb{G}_{0}^{(1,1;a)}(A_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z640}}$ $$\mathbb{Q}_{2,1}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z641}}$ $$\mathbb{Q}_{2,2}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z642}}$ $$\mathbb{Q}_{2,3}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z643}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g) = \frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z644}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g) = -\frac{\sqrt{6}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z645}}$ $$\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g) = -\frac{\sqrt{2}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z646}}$ $$\mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z647}}$ $$\mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z648}}$ $$\mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z649}}$ $$\mathbb{G}_{1,1}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z650}}$ $$\mathbb{G}_{1,2}^{(c)}(T_g) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z651}}$ $$\mathbb{G}_{1,3}^{(c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z652}}$ $$\mathbb{G}_{1,1}^{(1,-1;c)}(T_g) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10}$$

$\boxed{\text{z653}}$ $$\mathbb{G}_{1,2}^{(1,-1;c)}(T_g) = -\frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} - \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z654}}$ $$\mathbb{G}_{1,3}^{(1,-1;c)}(T_g) = \frac{\sqrt{30}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} + \frac{\sqrt{10}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z655}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,1) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{15}$$

$\boxed{\text{z656}}$ $$\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,1) = -\frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z657}}$ $$\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{5}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z741}}$ $$\mathbb{G}_{3,1}^{(1,-1;c)}(T_g,2) = -\frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z742}}$ $\mathbb{G}_{3,2}^{(1,-1;c)}(T_g,2) = \frac{\sqrt{3}\mathbb{M}_{2,1}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} - \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\mathbb{M}_{2,3}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$

$\boxed{\text{z743}}$ $\mathbb{G}_{3,3}^{(1,-1;c)}(T_g,2) = -\frac{\mathbb{M}_{2,1}^{(1,-1;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(E_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{M}_{2,2}^{(1,-1;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$

$\boxed{\text{z862}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z863}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g) = -\frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{T}_{1,3}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z864}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g) = \frac{\sqrt{6}\mathbb{T}_{1,1}^{(1,0;a)}(T_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{T}_{1,2}^{(1,0;a)}(T_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z865}}$ $\mathbb{G}_{1,1}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$

$\boxed{\text{z866}}$ $\mathbb{G}_{1,2}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$

$\boxed{\text{z867}}$ $\mathbb{G}_{1,3}^{(1,1;c)}(T_g) = \frac{\sqrt{3}\mathbb{M}_{0}^{(1,1;a)}(A_u)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z1357}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1358}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1359}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1360}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1361}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1362}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u,b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1363}}$ $\mathbb{Q}_{3,1}^{(c)}(T_u,2) = -\frac{\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1364}}$ $\mathbb{Q}_{3,2}^{(c)}(T_u,2) = \frac{\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1365}}$ $\mathbb{Q}_{3,3}^{(c)}(T_u,2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1366}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1367}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1368}}$ $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = -\frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1369}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1370}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1371}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1372}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1373}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1374}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,a) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1375}}$ $\mathbb{Q}_{1,1}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1376}}$ $\mathbb{Q}_{1,2}^{(1,0;c)}(T_u,b) = -\frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z1377}}$ $\mathbb{Q}_{1,3}^{(1,0;c)}(T_u,b) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1378}}$ $\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2) = -\frac{\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,1}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1379}}$ $\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2) = \frac{\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{1,2}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z1380}}$ $\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{Q}_{1,3}^{(1,0;a)}(T_u)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z1381}}$ $\mathbb{G}_{2,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,1}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1382}}$ $\mathbb{G}_{2,2}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,2}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z1383}}$ $\mathbb{G}_{2,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{3}\mathbb{G}_{2,3}^{(1,-1;a)}(T_u)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

- `A;A_002_3` : `'A'-'A'` bond-cluster
  * bra: $\langle p_x,\uparrow|$, $\langle p_x,\downarrow|$, $\langle p_y,\uparrow|$, $\langle p_y,\downarrow|$, $\langle p_z,\uparrow|$, $\langle p_z,\downarrow|$
  * ket: $|p_x,\uparrow\rangle$, $|p_x,\downarrow\rangle$, $|p_y,\uparrow\rangle$, $|p_y,\downarrow\rangle$, $|p_z,\uparrow\rangle$, $|p_z,\downarrow\rangle$
  * wyckoff: `3c@3c`

$\boxed{\text{z64}}$ $\mathbb{Q}_{0}^{(c)}(A_g,a) = \mathbb{Q}_{0}^{(a)}(A_g)\mathbb{Q}_{0}^{(b)}(A_g)$

$\boxed{\text{z65}}$ $\mathbb{Q}_{0}^{(c)}(A_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z66}}$ $\mathbb{Q}_{0}^{(1,-1;c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} + \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z67}}$ $\mathbb{Q}_{0}^{(1,1;c)}(A_g) = \mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{0}^{(b)}(A_g)$

$\boxed{\text{z68}}$ $\mathbb{G}_{3}^{(c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z69}}$ $\mathbb{G}_3^{(1,-1;c)}(A_g) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z196}}$ $\mathbb{Q}_3^{(1,-1;c)}(A_u) = -\frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} - \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z197}}$ $\mathbb{Q}_3^{(1,0;c)}(A_u) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z198}}$ $\mathbb{G}_0^{(c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z199}}$ $\mathbb{G}_0^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z200}}$ $\mathbb{G}_4^{(1,-1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z201}}$ $\mathbb{G}_0^{(1,1;c)}(A_u) = \frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z202}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z203}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_0^{(a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z204}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$

$\boxed{\text{z205}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,b) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$

$\boxed{\text{z206}}$ $\mathbb{Q}_{2,1}^{(c)}(E_g,c) = \frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z207}}$ $\mathbb{Q}_{2,2}^{(c)}(E_g,c) = -\frac{\mathbb{Q}_{2,1}^{(a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$

$\boxed{\text{z658}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_0^{(b)}(A_g)}{2}$

$\boxed{\text{z659}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,a) = \frac{\sqrt{2}\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{0}^{(b)}(A_g)}{2}$$

$\boxed{\text{z660}}$ $$\mathbb{Q}_{2,1}^{(1,-1;c)}(E_g,b) = \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z661}}$ $$\mathbb{Q}_{2,2}^{(1,-1;c)}(E_g,b) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2} - \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z662}}$ $$\mathbb{Q}_{2,1}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2}$$

$\boxed{\text{z663}}$ $$\mathbb{Q}_{2,2}^{(1,1;c)}(E_g) = \frac{\sqrt{2}\mathbb{Q}_{0}^{(1,1;a)}(A_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$$

$\boxed{\text{z664}}$ $$\mathbb{G}_{2,1}^{(c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z665}}$ $$\mathbb{G}_{2,2}^{(c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z666}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,a) = -\frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{14}$$

$\boxed{\text{z667}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,a) = \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{42}$$

$\boxed{\text{z668}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(E_u,b) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z669}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(E_u,b) = \frac{\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$$

$\boxed{\text{z670}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(E_u) = -\frac{\sqrt{210}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} + \frac{3\sqrt{14}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{210}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} - \frac{3\sqrt{14}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{210}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{42}$$

$\boxed{\text{z671}}$ $\mathbb{G}_{4,2}^{(1,-1;c)}(E_u) = \frac{\sqrt{70}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} - \frac{\sqrt{70}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{42}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{42}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{14}$

$\boxed{\text{z672}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(E_u) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z673}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(E_u) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z674}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(E_u) = -\frac{\sqrt{3}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{3}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{3}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$

$\boxed{\text{z675}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(E_u) = \frac{\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{2} - \frac{\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{2}$

$\boxed{\text{z676}}$ $\mathbb{Q}_{2,1}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z677}}$ $\mathbb{Q}_{2,2}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z678}}$ $\mathbb{Q}_{2,3}^{(c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z679}}$ $\mathbb{Q}_{2,1}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z680}}$ $\mathbb{Q}_{2,2}^{(c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z681}}$ $\mathbb{Q}_{2,3}^{(c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z682}}$ $\mathbb{Q}_{4,1}^{(c)}(T_g,1) = -\frac{\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z683}}$ $\mathbb{Q}_{4,2}^{(c)}(T_g,1) = \frac{\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z684}}$ $\mathbb{Q}_{4,3}^{(c)}(T_g,1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z744}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z745}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z746}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z747}}$ $\mathbb{Q}_{2,1}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z748}}$ $\mathbb{Q}_{2,2}^{(1,-1;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z749}}$ $\mathbb{Q}_{2,3}^{(1,-1;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z868}}$ $\mathbb{Q}_{4,1}^{(1,-1;c)}(T_g,1) = -\frac{\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z869}}$ $\mathbb{Q}_{4,2}^{(1,-1;c)}(T_g,1) = \frac{\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z870}}$ $\mathbb{Q}_{4,3}^{(1,-1;c)}(T_g,1) = \frac{\sqrt{3}\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z871}}$ $\mathbb{Q}_{2,1}^{(1,0;c)}(T_g) = -\frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z872}}$ $\mathbb{Q}_{2,2}^{(1,0;c)}(T_g) = \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{2} - \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{6}$

$\boxed{\text{z873}}$ $\mathbb{Q}_{2,3}^{(1,0;c)}(T_g) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{3}$

$\boxed{\text{z874}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z875}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z876}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,a) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{0}^{(b)}(A_g)}{3}$

$\boxed{\text{z877}}$ $\mathbb{G}_{1,1}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} + \frac{\mathbb{G}_{1,1}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z878}}$ $\mathbb{G}_{1,2}^{(1,0;c)}(T_g,b) = -\frac{\sqrt{3}\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{6} - \frac{\mathbb{G}_{1,2}^{(1,0;a)}(T_g)\mathbb{Q}_{2,2}^{(b)}(E_g)}{2}$

$\boxed{\text{z879}}$ $\mathbb{G}_{1,3}^{(1,0;c)}(T_g,b) = \frac{\sqrt{3}\mathbb{G}_{1,3}^{(1,0;a)}(T_g)\mathbb{Q}_{2,1}^{(b)}(E_g)}{3}$

$\boxed{\text{z1384}}$ $\mathbb{Q}_{1,1}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1385}}$ $\mathbb{Q}_{1,2}^{(c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1386}}$ $\mathbb{Q}_{1,3}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1387}}$ $\mathbb{Q}_{1,1}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1388}}$ $\mathbb{Q}_{1,2}^{(1,-1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1389}}$ $\mathbb{Q}_{1,3}^{(1,-1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1390}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,1) = -\frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12}$

$\boxed{\text{z1391}}$ $\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,1) = \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$

$\boxed{\text{z1392}}$ $\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,1) = -\frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12}$

$\boxed{\text{z1393}}$ $\mathbb{Q}_{3,1}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$

$\boxed{\text{z1394}}$ $$\mathbb{Q}_{3,2}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1395}}$ $$\mathbb{Q}_{3,3}^{(1,-1;c)}(T_u,2) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} + \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} + \frac{\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1396}}$ $$\mathbb{Q}_{1,1}^{(1,0;c)}(T_u) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1397}}$ $$\mathbb{Q}_{1,2}^{(1,0;c)}(T_u) = -\frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{30} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{10} - \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1398}}$ $$\mathbb{Q}_{1,3}^{(1,0;c)}(T_u) = \frac{\sqrt{30}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} + \frac{\sqrt{10}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} + \frac{\sqrt{10}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10}$$

$\boxed{\text{z1399}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} + \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1400}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,1) = -\frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{10} - \frac{\sqrt{15}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1401}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,1) = \frac{\sqrt{5}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{5} - \frac{\sqrt{15}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{15} - \frac{\sqrt{15}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{15}$$

$\boxed{\text{z1402}}$ $$\mathbb{Q}_{3,1}^{(1,0;c)}(T_u,2) = -\frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1403}}$ $$\mathbb{Q}_{3,2}^{(1,0;c)}(T_u,2) = \frac{\sqrt{3}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} - \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1404}}$ $$\mathbb{Q}_{3,3}^{(1,0;c)}(T_u,2) = -\frac{\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{3}$$

$\boxed{\text{z1405}}$ $$\mathbb{Q}_{1,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1406}}$ $$\mathbb{Q}_{1,2}^{(1,1;c)}(T_u) = -\frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1407}}$ $$\mathbb{Q}_{1,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1408}}$ $$\mathbb{G}_{2,1}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1409}}$ $$\mathbb{G}_{2,2}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1410}}$ $$\mathbb{G}_{2,3}^{(c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1411}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1412}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1413}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,a) = -\frac{\sqrt{21}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{21} - \frac{\sqrt{21}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} - \frac{\sqrt{35}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{21} + \frac{\sqrt{35}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{21}$$

$\boxed{\text{z1414}}$ $$\mathbb{G}_{2,1}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1415}}$ $$\mathbb{G}_{2,2}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1416}}$ $$\mathbb{G}_{2,3}^{(1,-1;c)}(T_u,b) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,-1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,-1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$$

$\boxed{\text{z1417}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,1) = \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1418}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,1) = -\frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{4} + \frac{\sqrt{15}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1419}}$ $$\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,1) = \frac{\sqrt{15}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{4} - \frac{\sqrt{15}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{12} - \frac{\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{4}$$

$\boxed{\text{z1420}}$ $$\mathbb{G}_{4,1}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{7}$$

$\boxed{\text{z1421}}$ $$\mathbb{G}_{4,2}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,3}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,3}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{7}$$

$\boxed{\text{z1422}}$ $\mathbb{G}_{4,3}^{(1,-1;c)}(T_u,2) = -\frac{\sqrt{105}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,2}^{(b)}(T_u)}{84} - \frac{3\sqrt{7}\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,2}^{(b)}(T_u)}{28} - \frac{\sqrt{105}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1)\mathbb{T}_{1,1}^{(b)}(T_u)}{84} + \frac{3\sqrt{7}\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2)\mathbb{T}_{1,1}^{(b)}(T_u)}{28} + \frac{\sqrt{7}\mathbb{M}_{3}^{(1,-1;a)}(A_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{7}$

$\boxed{\text{z1423}}$ $\mathbb{G}_{2,1}^{(1,0;c)}(T_u) = \frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1424}}$ $\mathbb{G}_{2,2}^{(1,0;c)}(T_u) = -\frac{\sqrt{6}\mathbb{T}_{2,1}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,3}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1425}}$ $\mathbb{G}_{2,3}^{(1,0;c)}(T_u) = -\frac{\sqrt{2}\mathbb{T}_{2,1}^{(1,0;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} - \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(E_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{3} + \frac{\sqrt{2}\mathbb{T}_{2,2}^{(1,0;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1426}}$ $\mathbb{G}_{2,1}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6}$

$\boxed{\text{z1427}}$ $\mathbb{G}_{2,2}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,3}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,3}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

$\boxed{\text{z1428}}$ $\mathbb{G}_{2,3}^{(1,1;c)}(T_u) = \frac{\sqrt{6}\mathbb{M}_{1,1}^{(1,1;a)}(T_g)\mathbb{T}_{1,2}^{(b)}(T_u)}{6} + \frac{\sqrt{6}\mathbb{M}_{1,2}^{(1,1;a)}(T_g)\mathbb{T}_{1,1}^{(b)}(T_u)}{6}$

Atomic SAMB

- bra: $\langle s,\uparrow|$, $\langle s,\downarrow|$
- ket: $|s,\uparrow\rangle$, $|s,\downarrow\rangle$

$\boxed{\text{x1}}$ $\mathbb{Q}_{0}^{(a)}(A_g) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & \frac{\sqrt{2}}{2} \end{bmatrix}$

$\boxed{\text{x2}}$ $\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & 0 \end{bmatrix}$

$\boxed{\text{x3}}$ $\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{2} \\ \frac{\sqrt{2}i}{2} & 0 \end{bmatrix}$

$\boxed{\text{x4}}$ $\mathbb{M}_{1,3}^{(1,-1;a)}(T_g) = \begin{bmatrix} \frac{\sqrt{2}}{2} & 0 \\ 0 & -\frac{\sqrt{2}}{2} \end{bmatrix}$

- bra: $\langle s, \uparrow |$, $\langle s, \downarrow |$
- ket: $|p_x, \uparrow\rangle$, $|p_x, \downarrow\rangle$, $|p_y, \uparrow\rangle$, $|p_y, \downarrow\rangle$, $|p_z, \uparrow\rangle$, $|p_z, \downarrow\rangle$

$\boxed{\text{x5}}$ $\mathbb{Q}_{1,1}^{(a)}(T_u) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x6}}$ $\mathbb{Q}_{1,2}^{(a)}(T_u) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x7}}$ $\mathbb{Q}_{1,3}^{(a)}(T_u) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$

$\boxed{\text{x8}}$ $\mathbb{Q}_{1,1}^{(1,0;a)}(T_u) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x9}}$ $\mathbb{Q}_{1,2}^{(1,0;a)}(T_u) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x10}}$ $\mathbb{Q}_{1,3}^{(1,0;a)}(T_u) = \begin{bmatrix} 0 & -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x11}}$ $\mathbb{G}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{12} & 0 & -\frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \end{bmatrix}$

$\boxed{\text{x12}}$ $\mathbb{G}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x13}}$ $\mathbb{G}_{2,1}^{(1,-1;a)}(T_u) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x14}}$ $\mathbb{G}_{2,2}^{(1,-1;a)}(T_u) = \begin{bmatrix} \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x15}}$ $\mathbb{G}_{2,3}^{(1,-1;a)}(T_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x16}}$ $\mathbb{G}_{0}^{(1,1;a)}(A_u) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \end{bmatrix}$

$\boxed{\text{x17}}$ $\mathbb{M}_{2,1}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{6} & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$

$\boxed{\text{x18}}$ $\mathbb{M}_{2,2}^{(1,-1;a)}(E_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ \frac{\sqrt{2}}{4} & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x19}}$ $\mathbb{M}_{2,1}^{(1,-1;a)}(T_u) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x20}}$ $\mathbb{M}_{2,2}^{(1,-1;a)}(T_u) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x21}}$ $\mathbb{M}_{2,3}^{(1,-1;a)}(T_u) = \begin{bmatrix} 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x22}}$ $\mathbb{M}_{0}^{(1,1;a)}(A_u) = \begin{bmatrix} 0 & \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & \frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} \end{bmatrix}$

$\boxed{\text{x23}}$ $\mathbb{T}_{1,1}^{(a)}(T_u) = \begin{bmatrix} \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x24}}$ $\mathbb{T}_{1,2}^{(a)}(T_u) = \begin{bmatrix} 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x25}}$ $\mathbb{T}_{1,3}^{(a)}(T_u) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \end{bmatrix}$

$\boxed{\text{x26}}$ $\mathbb{T}_{1,1}^{(1,0;a)}(T_u) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & \frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \end{bmatrix}$

$\boxed{\text{x27}}$ $\mathbb{T}_{1,2}^{(1,0;a)}(T_u) = \begin{bmatrix} \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \end{bmatrix}$

$\boxed{\text{x28}}$ $\mathbb{T}_{1,3}^{(1,0;a)}(T_u) = \begin{bmatrix} 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

- bra: $\langle p_x, \uparrow|$, $\langle p_x, \downarrow|$, $\langle p_y, \uparrow|$, $\langle p_y, \downarrow|$, $\langle p_z, \uparrow|$, $\langle p_z, \downarrow|$
- ket: $|p_x, \uparrow\rangle$, $|p_x, \downarrow\rangle$, $|p_y, \uparrow\rangle$, $|p_y, \downarrow\rangle$, $|p_z, \uparrow\rangle$, $|p_z, \downarrow\rangle$

$\boxed{\text{x29}}$ $\mathbb{Q}_{0}^{(a)}(A_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \end{bmatrix}$

$\boxed{\text{x30}}$ $\mathbb{Q}_{2,1}^{(a)}(E_g) = \begin{bmatrix} -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{3} \end{bmatrix}$

$\boxed{\text{x31}}$ $\mathbb{Q}_{2,2}^{(a)}(E_g) = \begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x32}}$ $\mathbb{Q}_{2,1}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x33}}$ $\mathbb{Q}_{2,2}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x34}}$ $\mathbb{Q}_{2,3}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & \frac{1}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x35}}$ $\mathbb{Q}_{2,1}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & \frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{12} \\ 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & \frac{\sqrt{6}i}{12} & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x36}}$ $\mathbb{Q}_{2,2}^{(1,-1;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & \frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x37}}$ $\mathbb{Q}_{2,1}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x38}}$ $\mathbb{Q}_{2,2}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x39}}$ $\mathbb{Q}_{2,3}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x40}}$ $\mathbb{Q}_{0}^{(1,1;a)}(A_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ \frac{\sqrt{3}i}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x41}}$ $\mathbb{G}_{1,1}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 \\ -\frac{\sqrt{2}}{4} & 0 & 0 & 0 & 0 & 0 \\ \frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x42}}$ $\mathbb{G}_{1,2}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & \frac{\sqrt{2}i}{4} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x43}}$ $\mathbb{G}_{1,3}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}}{4} & 0 \\ 0 & -\frac{\sqrt{2}i}{4} & 0 & -\frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x44}}$ $\mathbb{M}_{1,1}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -\frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{i}{2} \\ 0 & 0 & \frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{i}{2} & 0 & 0 \end{bmatrix}$

$\boxed{\text{x45}}$ $\mathbb{M}_{1,2}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & \frac{i}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{i}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -\frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{i}{2} & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x46}}$ $\mathbb{M}_{1,3}^{(a)}(T_g) = \begin{bmatrix} 0 & 0 & -\frac{i}{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{i}{2} & 0 & 0 \\ \frac{i}{2} & 0 & 0 & 0 & 0 & 0 \\ 0 & \frac{i}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x47}}$ $\mathbb{M}_3^{(1,-1;a)}(A_g) = \begin{bmatrix} 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \\ \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x48}}$ $\mathbb{M}_{1,1}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

$\boxed{\text{x49}}$ $\mathbb{M}_{1,2}^{(1,-1;a)}(T_g) = \begin{bmatrix} 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 \\ \frac{\sqrt{6}i}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}i}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

$\boxed{\text{x50}}$ $\mathbb{M}_{1,3}^{(1,-1;a)}(T_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{6} & 0 \\ 0 & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \end{bmatrix}$

$\boxed{\text{x51}}$ $\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,1) = \begin{bmatrix} 0 & \frac{\sqrt{5}}{5} & 0 & \frac{\sqrt{5}i}{10} & -\frac{\sqrt{5}}{10} & 0 \\ \frac{\sqrt{5}}{5} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \end{bmatrix}$

$\boxed{\text{x52}}$ $\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,1) = \begin{bmatrix} 0 & \frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 \\ -\frac{\sqrt{5}i}{10} & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{5} & -\frac{\sqrt{5}}{10} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{5} & 0 & 0 & \frac{\sqrt{5}}{10} \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \end{bmatrix}$

$\boxed{\text{x53}}$ $\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,1) = \begin{bmatrix} -\frac{\sqrt{5}}{10} & 0 & 0 & 0 & 0 & -\frac{\sqrt{5}}{10} \\ 0 & \frac{\sqrt{5}}{10} & 0 & 0 & -\frac{\sqrt{5}}{10} & 0 \\ 0 & 0 & -\frac{\sqrt{5}}{10} & 0 & 0 & \frac{\sqrt{5}i}{10} \\ 0 & 0 & 0 & \frac{\sqrt{5}}{10} & -\frac{\sqrt{5}i}{10} & 0 \\ 0 & -\frac{\sqrt{5}}{10} & 0 & \frac{\sqrt{5}i}{10} & \frac{\sqrt{5}}{5} & 0 \\ -\frac{\sqrt{5}}{10} & 0 & -\frac{\sqrt{5}i}{10} & 0 & 0 & -\frac{\sqrt{5}}{5} \end{bmatrix}$

$\boxed{\text{x54}}$ $\mathbb{M}_{3,1}^{(1,-1;a)}(T_g,2) = \begin{bmatrix} 0 & 0 & 0 & -\frac{\sqrt{3}i}{6} & -\frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 \\ \frac{\sqrt{3}i}{6} & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}}{6} & 0 \end{bmatrix}$

$\boxed{\text{x55}}$ $\mathbb{M}_{3,2}^{(1,-1;a)}(T_g,2) = \begin{bmatrix} 0 & \frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 \\ -\frac{\sqrt{3}i}{6} & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & 0 \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ -\frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{3}}{6} \\ 0 & 0 & \frac{\sqrt{3}}{6} & 0 & 0 & -\frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & -\frac{\sqrt{3}}{6} & \frac{\sqrt{3}i}{6} & 0 \end{bmatrix}$

$\boxed{\text{x56}}$ $\mathbb{M}_{3,3}^{(1,-1;a)}(T_g,2) = \begin{bmatrix} \frac{\sqrt{3}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{3}}{6} \\ 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}}{6} & 0 \\ 0 & 0 & -\frac{\sqrt{3}}{6} & 0 & 0 & \frac{\sqrt{3}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{3}}{6} & -\frac{\sqrt{3}i}{6} & 0 \\ 0 & \frac{\sqrt{3}}{6} & 0 & \frac{\sqrt{3}i}{6} & 0 & 0 \\ \frac{\sqrt{3}}{6} & 0 & -\frac{\sqrt{3}i}{6} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x57}}$ $\mathbb{M}_{1,1}^{(1,1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}}{15} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{15} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & -\frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 \\ \frac{\sqrt{30}i}{20} & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & 0 \\ \frac{\sqrt{30}}{20} & 0 & 0 & 0 & 0 & -\frac{\sqrt{30}}{30} \\ 0 & -\frac{\sqrt{30}}{20} & 0 & 0 & -\frac{\sqrt{30}}{30} & 0 \end{bmatrix}$

$\boxed{\text{x58}}$ $\mathbb{M}_{1,2}^{(1,1;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 \\ -\frac{\sqrt{30}i}{30} & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{15} & \frac{\sqrt{30}}{20} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{15} & 0 & 0 & -\frac{\sqrt{30}}{20} \\ 0 & 0 & \frac{\sqrt{30}}{20} & 0 & 0 & \frac{\sqrt{30}i}{30} \\ 0 & 0 & 0 & -\frac{\sqrt{30}}{20} & -\frac{\sqrt{30}i}{30} & 0 \end{bmatrix}$

$\boxed{\text{x59}}$ $\mathbb{M}_{1,3}^{(1,1;a)}(T_g) = \begin{bmatrix} -\frac{\sqrt{30}}{30} & 0 & 0 & 0 & 0 & \frac{\sqrt{30}}{20} \\ 0 & \frac{\sqrt{30}}{30} & 0 & 0 & \frac{\sqrt{30}}{20} & 0 \\ 0 & 0 & -\frac{\sqrt{30}}{30} & 0 & 0 & -\frac{\sqrt{30}i}{20} \\ 0 & 0 & 0 & \frac{\sqrt{30}}{30} & \frac{\sqrt{30}i}{20} & 0 \\ 0 & \frac{\sqrt{30}}{20} & 0 & -\frac{\sqrt{30}i}{20} & \frac{\sqrt{30}}{15} & 0 \\ \frac{\sqrt{30}}{20} & 0 & \frac{\sqrt{30}i}{20} & 0 & 0 & -\frac{\sqrt{30}}{15} \end{bmatrix}$

$\boxed{\text{x60}}$ $\mathbb{T}_{2,1}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}i}{4} \\ 0 & 0 & 0 & 0 & -\frac{\sqrt{2}i}{4} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} \\ 0 & 0 & 0 & 0 & \frac{\sqrt{2}}{4} & 0 \\ 0 & \frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 \\ -\frac{\sqrt{2}i}{4} & 0 & \frac{\sqrt{2}}{4} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x61}}$ $\mathbb{T}_{2,2}^{(1,0;a)}(E_g) = \begin{bmatrix} 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ -\frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}}{6} & 0 & 0 & \frac{\sqrt{6}}{12} & 0 \\ 0 & -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \end{bmatrix}$

$\boxed{\text{x62}}$ $\mathbb{T}_{2,1}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & 0 & 0 & \frac{\sqrt{6}i}{12} & \frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & \frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 \\ -\frac{\sqrt{6}i}{12} & 0 & \frac{\sqrt{6}}{6} & 0 & 0 & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{6} \\ 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{6} & 0 \end{bmatrix}$

$\boxed{\text{x63}}$ $\mathbb{T}_{2,2}^{(1,0;a)}(T_g) = \begin{bmatrix} 0 & \frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 \\ -\frac{\sqrt{6}i}{6} & 0 & \frac{\sqrt{6}}{12} & 0 & 0 & 0 \\ 0 & \frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ \frac{\sqrt{6}}{12} & 0 & 0 & 0 & 0 & \frac{\sqrt{6}}{12} \\ 0 & 0 & -\frac{\sqrt{6}}{12} & 0 & 0 & -\frac{\sqrt{6}i}{6} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{12} & \frac{\sqrt{6}i}{6} & 0 \end{bmatrix}$

$\boxed{\text{x64}}$ $\mathbb{T}_{2,3}^{(1,0;a)}(T_g) = \begin{bmatrix} \frac{\sqrt{6}}{6} & 0 & 0 & 0 & 0 & -\frac{\sqrt{6}}{12} \\ 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}}{12} & 0 \\ 0 & 0 & -\frac{\sqrt{6}}{6} & 0 & 0 & -\frac{\sqrt{6}i}{12} \\ 0 & 0 & 0 & \frac{\sqrt{6}}{6} & \frac{\sqrt{6}i}{12} & 0 \\ 0 & -\frac{\sqrt{6}}{12} & 0 & -\frac{\sqrt{6}i}{12} & 0 & 0 \\ -\frac{\sqrt{6}}{12} & 0 & \frac{\sqrt{6}i}{12} & 0 & 0 & 0 \end{bmatrix}$

Cluster SAMB

- Site cluster

** Wyckoff: `3d`

$\boxed{\text{y1}}$ $\mathbb{Q}_0^{(s)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

$\boxed{\text{y2}}$ $\mathbb{Q}_{2,1}^{(s)}(E_g) = \left[-\frac{10\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{2778}, \frac{43\sqrt{2778}}{2778}\right]$

$\boxed{\text{y3}}$ $\mathbb{Q}_{2,2}^{(s)}(E_g) = \left[\frac{11\sqrt{926}}{463}, -\frac{21\sqrt{926}}{926}, -\frac{\sqrt{926}}{926}\right]$

- Bond cluster

** Wyckoff: `3a@1a`

$\boxed{\text{y4}}$ $\mathbb{Q}_0^{(s)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

$\boxed{\text{y5}}$ $\mathbb{Q}_{2,1}^{(s)}(E_g) = \left[-\frac{10\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{2778}, \frac{43\sqrt{2778}}{2778}\right]$

$\boxed{\text{y6}}$ $\mathbb{Q}_{2,2}^{(s)}(E_g) = \left[\frac{11\sqrt{926}}{463}, -\frac{21\sqrt{926}}{926}, -\frac{\sqrt{926}}{926}\right]$

$\boxed{\text{y7}}$ $\mathbb{T}_{1,1}^{(s)}(T_u) = [i,\ 0,\ 0]$

$\boxed{\text{y8}}$ $\mathbb{T}_{1,2}^{(s)}(T_u) = [0,\ i,\ 0]$

$\boxed{\text{y9}}$ $\mathbb{T}_{1,3}^{(s)}(T_u) = [0,\ 0,\ i]$

** Wyckoff: `3b@3c`

$\boxed{\text{y10}}$ $\mathbb{Q}_{0}^{(s)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

$\boxed{\text{y11}}$ $\mathbb{Q}_{2,1}^{(s)}(E_g) = \left[-\frac{10\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{2778}, \frac{43\sqrt{2778}}{2778}\right]$

$\boxed{\text{y12}}$ $\mathbb{Q}_{2,2}^{(s)}(E_g) = \left[\frac{11\sqrt{926}}{463}, -\frac{21\sqrt{926}}{926}, -\frac{\sqrt{926}}{926}\right]$

$\boxed{\text{y13}}$ $\mathbb{T}_{1,1}^{(s)}(T_u) = [0,\ i,\ 0]$

$\boxed{\text{y14}}$ $\mathbb{T}_{1,2}^{(s)}(T_u) = [0,\ 0,\ i]$

$\boxed{\text{y15}}$ $\mathbb{T}_{1,3}^{(s)}(T_u) = [i,\ 0,\ 0]$

** Wyckoff: `3c@3c`

$\boxed{\text{y16}}$ $\mathbb{Q}_{0}^{(s)}(A_g) = \left[\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right]$

$\boxed{\text{y17}}$ $\mathbb{Q}_{2,1}^{(s)}(E_g) = \left[-\frac{10\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{2778}, \frac{43\sqrt{2778}}{2778}\right]$

$\boxed{\text{y18}}$ $\mathbb{Q}_{2,2}^{(s)}(E_g) = \left[\frac{11\sqrt{926}}{463}, -\frac{21\sqrt{926}}{926}, -\frac{\sqrt{926}}{926}\right]$

$\boxed{\text{y19}}$ $\mathbb{T}_{1,1}^{(s)}(T_u) = [0,\ 0,\ i]$

$\boxed{\text{y20}}$ $\mathbb{T}_{1,2}^{(s)}(T_u) = [i,\ 0,\ 0]$

$\boxed{\text{y21}}$ $\mathbb{T}_{1,3}^{(s)}(T_u) = [0,\ i,\ 0]$

** Wyckoff: `12a@12j`

$\boxed{\text{y22}}$ $\mathbb{Q}_{0}^{(s)}(A_g) = \left[\frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}, \frac{\sqrt{3}}{6}\right]$

$\boxed{\text{y23}}$ $\mathbb{T}_{0}^{(s)}(A_g) = \left[\frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}, \frac{\sqrt{3}i}{6}\right]$

$\boxed{\text{y24}}$ $\mathbb{Q}_{2,1}^{(s)}(E_g) = \left[-\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{5\sqrt{2778}}{1389}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, -\frac{23\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}, \frac{43\sqrt{2778}}{5556}\right]$

$\boxed{\text{y25}}$ $\mathbb{Q}_{2,2}^{(s)}(E_g) = \left[\frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, \frac{11\sqrt{926}}{926}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{21\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}, -\frac{\sqrt{926}}{1852}\right]$

$\boxed{\text{y26}}$ $\mathbb{T}_{2,1}^{(s)}(E_g) = \left[-\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{5\sqrt{2778}i}{1389}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, -\frac{23\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}, \frac{43\sqrt{2778}i}{5556}\right]$

$\boxed{\text{y27}}$ $\mathbb{T}_{2,2}^{(s)}(E_g) = \left[\frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, \frac{11\sqrt{926}i}{926}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{21\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}, -\frac{\sqrt{926}i}{1852}\right]$

$\boxed{\text{y28}}$ $\mathbb{M}_{1,1}^{(s)}(T_g) = \left[\frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, 0, 0, 0, 0, 0, 0, 0, 0\right]$

$\boxed{\text{y29}}$ $\mathbb{M}_{1,2}^{(s)}(T_g) = \left[0, 0, 0, 0, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}, 0, 0, 0, 0\right]$

$\boxed{\text{y30}}$ $\mathbb{M}_{1,3}^{(s)}(T_g) = \left[0, 0, 0, 0, 0, 0, 0, 0, \frac{i}{2}, -\frac{i}{2}, -\frac{i}{2}, \frac{i}{2}\right]$

$\boxed{\text{y31}}$ $\mathbb{Q}^{(s)}_{2,1}(T_g) = \left[\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0, 0, 0, 0, 0\right]$

$\boxed{\text{y32}}$ $\mathbb{Q}^{(s)}_{2,2}(T_g) = \left[0, 0, 0, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, 0, 0, 0, 0\right]$

$\boxed{\text{y33}}$ $\mathbb{Q}^{(s)}_{2,3}(T_g) = \left[0, 0, 0, 0, 0, 0, 0, 0, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}\right]$

$\boxed{\text{y34}}$ $\mathbb{Q}^{(s)}_{1,1}(T_u) = \left[0, 0, 0, 0, \frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}\right]$

$\boxed{\text{y35}}$ $\mathbb{Q}^{(s)}_{1,2}(T_u) = \left[\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, 0, 0, 0, 0, \frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}\right]$

$\boxed{\text{y36}}$ $\mathbb{Q}^{(s)}_{1,3}(T_u) = \left[\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, 0, 0, 0, 0\right]$

$\boxed{\text{y37}}$ $\mathbb{T}^{(s)}_{1,1}(T_u, a) = \left[0, 0, 0, 0, \frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}\right]$

$\boxed{\text{y38}}$ $\mathbb{T}^{(s)}_{1,2}(T_u, a) = \left[\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, 0, 0, 0, 0, \frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}\right]$

$\boxed{\text{y39}}$ $\mathbb{T}^{(s)}_{1,3}(T_u, a) = \left[\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, 0, 0, 0, 0\right]$

$\boxed{\text{y40}}$ $\mathbb{T}^{(s)}_{1,1}(T_u, b) = \left[0, 0, 0, 0, \frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}\right]$

$\boxed{\text{y41}}$ $\mathbb{T}^{(s)}_{1,2}(T_u, b) = \left[-\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, 0, 0, 0, 0, \frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}\right]$

$\boxed{\text{y42}}$ $\mathbb{T}^{(s)}_{1,3}(T_u, b) = \left[\frac{\sqrt{21}i}{14}, \frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{21}i}{14}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, -\frac{\sqrt{7}i}{7}, \frac{\sqrt{7}i}{7}, 0, 0, 0, 0\right]$

$\boxed{\text{y43}}$ $\mathbb{Q}^{(s)}_{3,1}(T_u, 1) = \left[0, 0, 0, 0, \frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}\right]$

$\boxed{\text{y44}}$ $\mathbb{Q}^{(s)}_{3,2}(T_u, 1) = \left[-\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, 0, 0, 0, 0, \frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}\right]$

$\boxed{\text{y45}}$ $\mathbb{Q}^{(s)}_{3,3}(T_u, 1) = \left[\frac{\sqrt{21}}{14}, \frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{21}}{14}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, -\frac{\sqrt{7}}{7}, \frac{\sqrt{7}}{7}, 0, 0, 0, 0\right]$

Site and Bond

Table 5: Orbital of each site

#	site	orbital
1	`A`	$\lvert s,\uparrow\rangle$, $\lvert s,\downarrow\rangle$, $\lvert p_x,\uparrow\rangle$, $\lvert p_x,\downarrow\rangle$, $\lvert p_y,\uparrow\rangle$, $\lvert p_y,\downarrow\rangle$, $\lvert p_z,\uparrow\rangle$, $\lvert p_z,\downarrow\rangle$

Table 6: Neighbor and bra-ket of each bond

#	head	tail	neighbor	head (bra)	tail (ket)
1	`A`	`A`	`[1,2]`	`[s,p]`	`[s,p]`

Site in Unit Cell

Sites in (conventional) cell (no plus set), SL = sublattice

Table 7: 'A' (#1) site cluster (3d), mmm..

SL	position ($\boldsymbol{s}$)	mapping
1	[0.50000, 0.00000, 0.00000]	[1,2,3,4,13,14,15,16]
2	[0.00000, 0.50000, 0.00000]	[5,6,7,8,17,18,19,20]
3	[0.00000, 0.00000, 0.50000]	[9,10,11,12,21,22,23,24]

Bond in Unit Cell

Bonds in (conventional) cell (no plus set): tail, head = (SL, plus set), (N)D = (non)directional (listed up to 5th neighbor at most)

Table 8: 1-th 'A'-'A' [1] (#1) bond cluster (12a@12j), D, $|\boldsymbol{v}|$= 0.70711 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000,-0.50000,-0.50000]	[0.00000, 0.25000, 0.75000]	[1,16]	(3,1)	(2,1)	[0,0,1]
2	[0.00000, 0.50000,-0.50000]	[0.00000, 0.75000, 0.75000]	[2,15]	(3,1)	(2,1)	[0,-1,1]
3	[0.00000,-0.50000, 0.50000]	[0.00000, 0.25000, 0.25000]	[3,14]	(3,1)	(2,1)	[0,0,0]
4	[0.00000, 0.50000, 0.50000]	[0.00000, 0.75000, 0.25000]	[4,13]	(3,1)	(2,1)	[0,-1,0]
5	[-0.50000, 0.00000,-0.50000]	[0.75000, 0.00000, 0.25000]	[5,20]	(1,1)	(3,1)	[1,0,0]
6	[-0.50000, 0.00000, 0.50000]	[0.75000, 0.00000, 0.75000]	[6,19]	(1,1)	(3,1)	[1,0,-1]
7	[0.50000, 0.00000,-0.50000]	[0.25000, 0.00000, 0.25000]	[7,18]	(1,1)	(3,1)	[0,0,0]
8	[0.50000, 0.00000, 0.50000]	[0.25000, 0.00000, 0.75000]	[8,17]	(1,1)	(3,1)	[0,0,-1]
9	[-0.50000,-0.50000, 0.00000]	[0.25000, 0.75000, 0.00000]	[9,24]	(2,1)	(1,1)	[0,1,0]

continued ...

Table 8

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
10	[0.50000,-0.50000, 0.00000]	[0.75000, 0.75000, 0.00000]	[10,23]	(2,1)	(1,1)	[-1,1,0]
11	[-0.50000, 0.50000, 0.00000]	[0.25000, 0.25000, 0.00000]	[11,22]	(2,1)	(1,1)	[0,0,0]
12	[0.50000, 0.50000, 0.00000]	[0.75000, 0.25000, 0.00000]	[12,21]	(2,1)	(1,1)	[-1,0,0]

Table 9: 2-th 'A'-'A' [1] (#2) bond cluster (3a@1a), ND, $|\boldsymbol{v}|= 1.0$ (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[-1.00000, 0.00000, 0.00000]	[0.00000, 0.00000, 0.00000]	[1,-2,-3,4,-13,14,15,-16]	(1,1)	(1,1)	[1,0,0]
2	[0.00000,-1.00000, 0.00000]	[0.00000, 0.00000, 0.00000]	[5,-6,-7,8,-17,18,19,-20]	(2,1)	(2,1)	[0,1,0]
3	[0.00000, 0.00000,-1.00000]	[0.00000, 0.00000, 0.00000]	[9,-10,-11,12,-21,22,23,-24]	(3,1)	(3,1)	[0,0,1]

Table 10: 2-th 'A'-'A' [2] (#3) bond cluster (3b@3c), ND, $|\boldsymbol{v}|= 1.0$ (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000, 0.00000,-1.00000]	[0.00000, 0.50000, 0.50000]	[1,2,-3,-4,-13,-14,15,16]	(2,1)	(2,1)	[0,0,1]
2	[-1.00000, 0.00000, 0.00000]	[0.50000, 0.00000, 0.50000]	[5,6,-7,-8,-17,-18,19,20]	(3,1)	(3,1)	[1,0,0]
3	[0.00000,-1.00000, 0.00000]	[0.50000, 0.50000, 0.00000]	[9,10,-11,-12,-21,-22,23,24]	(1,1)	(1,1)	[0,1,0]

Table 11: 2-th 'A'-'A' [3] (#4) bond cluster (3c@3c), ND, $|\boldsymbol{v}|$= 1.0 (cartesian)

SL	vector ($\boldsymbol{v}$)	center ($\boldsymbol{c}$)	mapping	head	tail	$\boldsymbol{R}$ (primitive)
1	[0.00000,-1.00000, 0.00000]	[0.00000, 0.50000, 0.50000]	[1,-2,3,-4,-13,14,-15,16]	(3,1)	(3,1)	[0,1,0]
2	[0.00000, 0.00000,-1.00000]	[0.50000, 0.00000, 0.50000]	[5,-6,7,-8,-17,18,-19,20]	(1,1)	(1,1)	[0,0,1]
3	[-1.00000, 0.00000, 0.00000]	[0.50000, 0.50000, 0.00000]	[9,-10,11,-12,-21,22,-23,24]	(2,1)	(2,1)	[1,0,0]